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ORIGINAL ARTICLE

Neocortical neuron types in Xenarthra and Afrotheria: implications for brain evolution in mammals

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Abstract Interpreting the evolution of neuronal types in the cerebral cortex of mammals requires information from a diversity of species. However, there is currently a paucity of data from the Xenarthra and Afrotheria, two major phylogenetic groups that diverged close to the base of the eutherian mammal adaptive radiation. In this study, we used immunohistochemistry to examine the distribution and morphology of neocortical neurons stained for non-phosphorylated neurofilament protein, calbindin, calretinin, parvalbumin, and neuropeptide Y in three xenarthran species—the giant anteater (*Myrmecophaga tridactyla*), the lesser anteater (*Tamandua tetradactyla*), and the two-toed sloth (*Choloepus didactylus*)—and two afrotherian species—the rock hyrax (*Procavia capensis*) and the black and rufous giant elephant shrew (*Rhynchocyon petersi*). We also studied the distribution and morphology of astrocytes using glial fibrillary acidic protein as a marker. In all of these species, nonphosphorylated neurofilament protein-immunoreactive neurons predominated in layer V. These neurons exhibited diverse morphologies with regional variation. Specifically, high proportions of atypical neurofilament-enriched neuron classes were observed, including extraverted neurons, inverted pyramidal neurons, fusiform neurons, and other multipolar types. In addition, many projection neurons in layers II–III were found to contain calbindin. Among interneurons, parvalbumin- and calbindin-expressing cells were generally denser compared to calretinin-immunoreactive cells. We traced the evolution of certain cortical architectural traits using phylogenetic analysis. Based on our reconstruction of character evolution, we found that the living xenarthrans and afrotherians show many similarities to the stem eutherian mammal, whereas other eutherian lineages display a greater number of derived traits.

Keywords Brain evolution · Cerebral cortex · Interneuron · Mammal · Pyramidal cell

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Introduction

A complete view of mammalian brain evolution requires information from a wide diversity of species encompassing key branching points in the phylogenetic tree (Bullock 1984; Johnson et al. 1984; Kaas 2006). According to molecular genetic studies, the living eutherian (placental) mammals are divided into four major groups including Afrotheria,

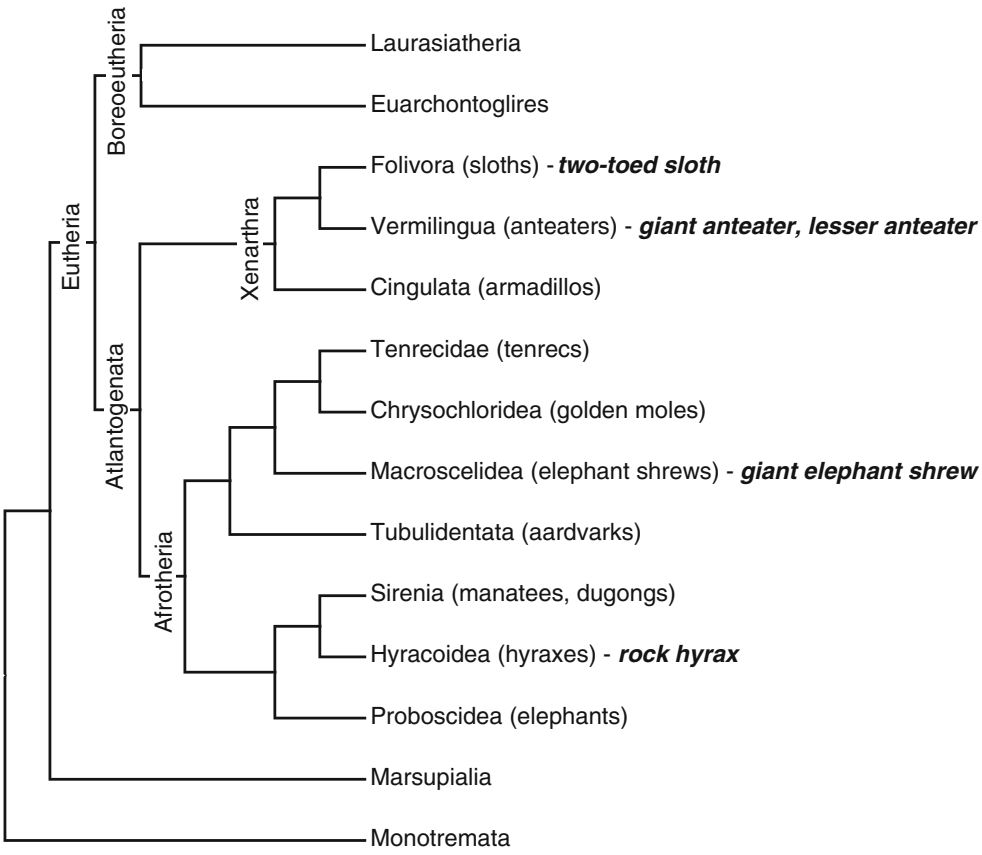


Fig. 1 Phylogenetic tree of mammals, showing the position of species included in the current study

Xenarthra, Euarchontoglires, and Laurasiatheria (Murphy et al. 2007; Wildman et al. 2007). Fossil evidence suggests that these distinct clades were well established prior to the K-T boundary at 65 million years ago (Wible et al. 2007), with xenarthrans and afrotherians arising on the Southern Hemisphere supercontinent of Gondwana and euarchontoglires and laurasiatherians originating on the Northern Hemisphere supercontinent of Laurasia.

A considerable amount is known about the cyto- and chemoarchitecture of the neocortex in many different mammalian taxa, including prototherians (monotremes—e.g., echidna), metatherians (marsupials—e.g., wallaby, opossum), and the eutherian euarchontoglires (e.g., rodents, primates) and laurasiatherians (e.g., carnivores, chiropterans, cetartiodactyls) (Hassiotis and Ashwell 2003; Hassiotis et al. 2003, 2004, 2005; Hof et al. 1999, 2000; Hof and Sherwood 2005). Among these mammals, pronounced phylogenetic variation has been observed in the morphology, distribution, and protein expression of neocortical cell types as revealed by immunostaining against nonphosphorylated epitopes on the neurofilament triplet protein (NPNFP), calcium-binding proteins, neuropeptides, and glial fibrillary acidic protein (GFAP) (Ballesteros-Yañez et al. 2005; Colombo et al. 2000; DeFelipe et al. 2002; Glezer et al. 1993; Hassiotis et al. 2005; Hof and Sherwood 2005; Preuss 2000).

In contrast, xenarthrans and afrotherians are drastically underrepresented in comparative neuroanatomical studies. Because xenarthrans and afrotherians are joined on one common branch at the base of the eutherian radiation within the Atlantogenata (Fig. 1), they occupy a pivotal phylogenetic position to shed light on the evolution of neocortical organization in mammals. Therefore, to provide a more comprehensive reconstruction of the evolution of brain cell types, in this study we characterized the morphological and biochemical phenotype of neocortical neurons and astroglia in representative xenarthran and afrotherian species.

The Xenarthra is a New World lineage united by shared derived traits such as the absence of incisors and canines, postcranial adaptations for digging and burrowing, an especially low metabolic rate, and variable regulation of body temperature (Vizcaíno and Loughry 2008). Living members of the Xenarthra comprise animals with insectivorous and herbivorous diets, including species of anteaters, sloths, and armadillos. Today, xenarthrans are found mostly in Central and South America, with only the nine-banded armadillo's range extending to North America. The extant species of Xenartha, however, represent only a small fraction of the past diversity and distribution of this group. In the Tertiary period, from 65 to 1.8 million years ago, xenarthrans were considerably more numerous,

radiating into more than 150 different genera spread throughout the New World. Paleontological evidence demonstrates that some extinct xenarthrans, such as the giant armored *Glyptodon* and the giant ground sloth *Megatherium*, attained enormous body sizes.

The Afrotheria is composed of six orders that arose in Africa and whose living members are still largely found in Afro-Arabia. Extant afrotherian species are highly diversified and include elephants, manatees, dugongs, hyraxes, aardvarks, golden moles, tenrecs, and elephant shrews. This clade represents an impressive range of variation in terms of behavior, brain size, and body size. Because the living afrotherian orders are thought to represent the tips of very long evolutionary branches that extend back to the Late Cretaceous (Seiffert 2007), extreme divergent adaptations characterize the crown members of this clade. Therefore, despite strong support for the monophyly of Afrotheria by molecular data, few morphological synapomorphies have been found that unite this group (Asher and Lehmann 2008; Carter et al. 2006; Sanchez-Villagra et al. 2007).

Previous investigations of neocortical organization in xenarthrans and afrotherians include electrophysiological studies of functional localization in the nine-banded armadillo (Dom et al. 1971; Royce et al. 1975), two-toed sloth (Meulders et al. 1966), three-toed sloth (De Moraes et al. 1963; Saraiva and Magalhães-Castro 1975), Madagascan lesser hedgehog tenrec (Krubitzer et al. 1997), and Cape elephant shrew (Dengler-Crish et al. 2006). In addition, there are reports of cortical architecture and parcellation in armadillos (Dom et al. 1971; Ferrari et al. 1998; Royce et al. 1975), the two-toed sloth (Gerebtzoff and Goffart 1966), Madagascan lesser hedgehog tenrec (Schmolke and Künzle 1997), manatee (Marshall and Reep 1995; Reep et al. 1989; Sarko and Reep 2007), and elephant (Cozzi et al. 2001). None of these studies, however, provide detailed descriptions of the distribution of cellular types in the neocortex as defined by immunohistochemical staining patterns.

In this study, we used immunohistochemistry to characterize the cell types of the neocortex in three species of xenarthrans—the giant anteater (*Myrmecophaga tridactyla*), the lesser anteater (*Tamandua tetradactyla*), and the two-toed sloth (*Choloepus didactylus*)—and two species of afrotherians—the rock hyrax (*Procavia capensis*) and the black and rufous giant elephant shrew (*Rhynchocyon petersi*). We examined whether the chemoarchitectural organization of neocortex accords with phylogenetic relations among these taxa and we used these data to reconstruct the evolution of neocortical neuron types among mammals.

Materials and methods

Specimens

Details about brain specimens used in this study are presented in Table 1 and lateral views are shown in Fig. 2. All

Table 1 Specimens used in the study

Species	Common name	Sex	Age	Brain mass (g)	Body mass (kg)	Length of fixation (days)
Choloepus didactylus	Two-toed sloth	F	Unknown > 8 years	28	3.2	17
Choloepus didactylus	Two-toed sloth	F	2 years, 11 m	36	7.7	10
Tamandua tetradactyla	Lesser anteater	F	Unknown > 6 years	30	7.0	10
Myrmecophaga tridactyla	Giant anteater	F	Unknown > 1 year, 4 month	76[a]	52.1	7
Procavia capensis	Rock hyrax	M	1 year, 2 month	19	1.8	7
Procavia capensis	Rock hyrax	F	1 year, 5 month	18	3.6	7
Rhynchocyon petersi	Giant elephant shrew	F	5 year, 0 month	5.4	0.471	10

[a] The actual brain mass of this specimen was not available. The value presented here is taken from the volumetric data in Bush and Allman (2004) and adjusted for the specific gravity of brain tissue (1.036 g/cm^3)

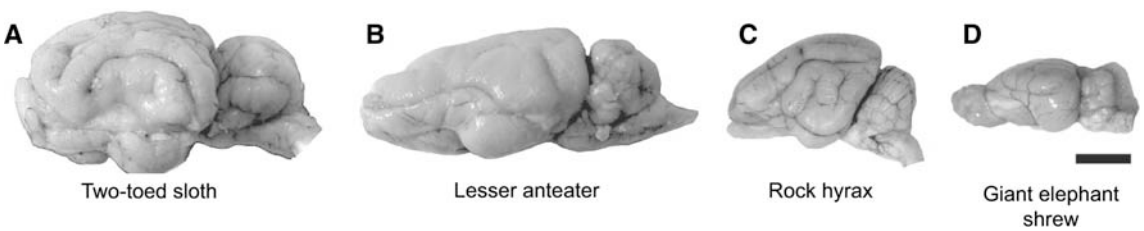


Fig. 2 Left lateral views of brains used in this study: **a** two-toed sloth, **b** lesser anteater, **c** rock hyrax, and **d** giant elephant shrew. The giant anteater brain that was obtained for this study had previously been dissected for pathology examination and so is not illustrated. *Scale bar* 1 cm

brains except the giant elephant shrew were obtained from the Cleveland Metroparks Zoo. The giant elephant shrew brain came from the Philadelphia Zoo. Considering the species-specific age of sexual maturity, the giant anteater and one of the two-toed sloths were juveniles; all other specimens were from adults. Samples were obtained post-mortem after animals died for reasons unrelated to the current study. All specimens appeared normal on routine pathology examination. Brains were removed at the zoo veterinary hospital within 14 h after death and were fixed by immersion in 10% buffered formalin. Brains remained in formalin solution between 7 and 17 days, and then were stored in phosphate buffered saline (PBS) with 0.1% sodium azide at 4°C until sectioning. In all cases except the giant anteater specimen, which had previously been blocked and sampled for pathology examination, brain weights were measured either at the zoo or immediately upon receipt in the laboratory. For comparative purposes, we used a brain mass of 76 g for the giant anteater, provided in Bush and Allman (2004).

Histological preparation and immunohistochemistry

For most specimens, the entire left cerebral hemisphere of each brain was sectioned for histology. The hindbrain was separated from the cerebrum by cutting at the level of the substantia nigra and the hemispheres were separated from each other midsagittally. All available tissue blocks from the giant anteater's left cerebral hemisphere were histologically processed. Several coronal blocks of the giant elephant shrew's right hemisphere were made available for this study. Brain tissue was cryoprotected by immersion with increasing concentrations of sucrose solution up to 30%. The samples were then frozen on a slur of dry ice and isopentane, embedded in tissue freezing medium, and sections were cut at 40 µm using a sliding microtome. From each specimen, a 1:10 series of sections was stained for Nissl substance with a solution of 0.5% cresyl violet. In addition, a 1:20 series was stained for myelin with the Gallyas (1979) method.

Immunohistochemistry was performed for each antigen on adjacent 1:20 series of sections. Free-floating sections were stained with monoclonal antibodies against nonphosphorylated epitopes on the neurofilament protein triplet (NPNFP; dilution 1:3,000; SMI-32 antibody; Covance International, Princeton, NJ, USA), neuropeptide Y (NPY; dilution 1:1,500; Peninsula Laboratories, Inc., San Carlos, CA, USA), glial fibrillary acidic protein (GFAP; dilution 1:7,500; Dako, Glostrup, Denmark), parvalbumin (PV; dilution 1:10,000; Swant, Bellinzona, Switzerland) and calbindin D-28k (CB; dilution 1:8,000; Swant), and with a polyclonal antibody against calretinin (CR; dilution 1:10,000; Swant). Prior to immunostaining, sections were rinsed thoroughly in PBS, pretreated for antigen retrieval by incubation in 10 mM sodium citrate buffer (pH 3.5) at 37°C in an oven for 30 min, then immersed in a solution of 0.75% hydrogen peroxide in 75% methanol to eliminate endogenous peroxidase activity. For the SMI-32 antibody directed against NPNFP, antigen retrieval used the same buffer with pH 8.5 at 85°C in a water bath to achieve optimal staining results. Primary antibodies were diluted in a solution containing PBS with 2% normal serum (4% normal serum for SMI-32) and 0.1% Triton X-100 and incubated for approximately 48 h on a rotating table at 4°C. After rinsing in PBS, sections were incubated in the biotinylated secondary IgG antibody (dilution 1:200, Vector Laboratories, Burlingame, CA, USA) and processed with the avidin-biotin-peroxidase method using a Vectastain Elite ABC kit (Vector Laboratories). Immunoreactivity was revealed using 3,3'-diaminobenzidine (Vector Laboratories). Nickel enhancement of the chromogen was used in some cases. Negative control sections were processed as described above excluding the primary antibody. Immunostaining was completely absent in control sections. Positive control sections of 4% paraformaldehyde-perfused mouse cerebral cortex tissue were also used in all experiments. Alternate immunoreacted sections were counterstained with cresyl violet to facilitate identification of layers and regional boundaries for analyses of cell densities.

Regional and laminar analysis

The regional and laminar analysis of immunostaining results was performed using a Zeiss Axioplan 2 photomicroscope equipped with a Ludl XY motorized stage (Ludl Electronics, Hawthorne, NY, USA), Heidenhain *z*-axis encoder (Heidenhain Corp., Schaumburg, IL, USA), an Optronics MicroFire color videocamera (Optronics, Goleta, CA, USA), and a Dell PC workstation running the StereoInvestigator and Neurolucida software (MBF Bioscience, Williston, VT, USA). We identified cortical regions for examination based on location and architecture observed in sections stained for Nissl, myelin, NPNFP, and PV. In addition, we referred to published atlases and cortical maps of echidna (Hassiotis et al. 2004), nine-banded armadillo (Royce et al. 1975), tammar wallaby (Ashwell et al. 2005), Madagascan lesser hedgehog tenrec (Krubitzer et al. 1997), Cape elephant shrew (Dengler-Crish et al. 2006), and rat (Paxinos and Watson 2005). For this study, we did not perform a detailed areal parcellation of the cerebral cortex in these species. Nonetheless, the general layout of the neocortex was determined with respect to the location of primary sensory areas, which could be distinguished by heavy myelination and intense PV staining of the neuropil in layer IV (Fig. 3).

Illustrations of neuronal morphologies in the most dorsal portion of the primary somatosensory cortex were

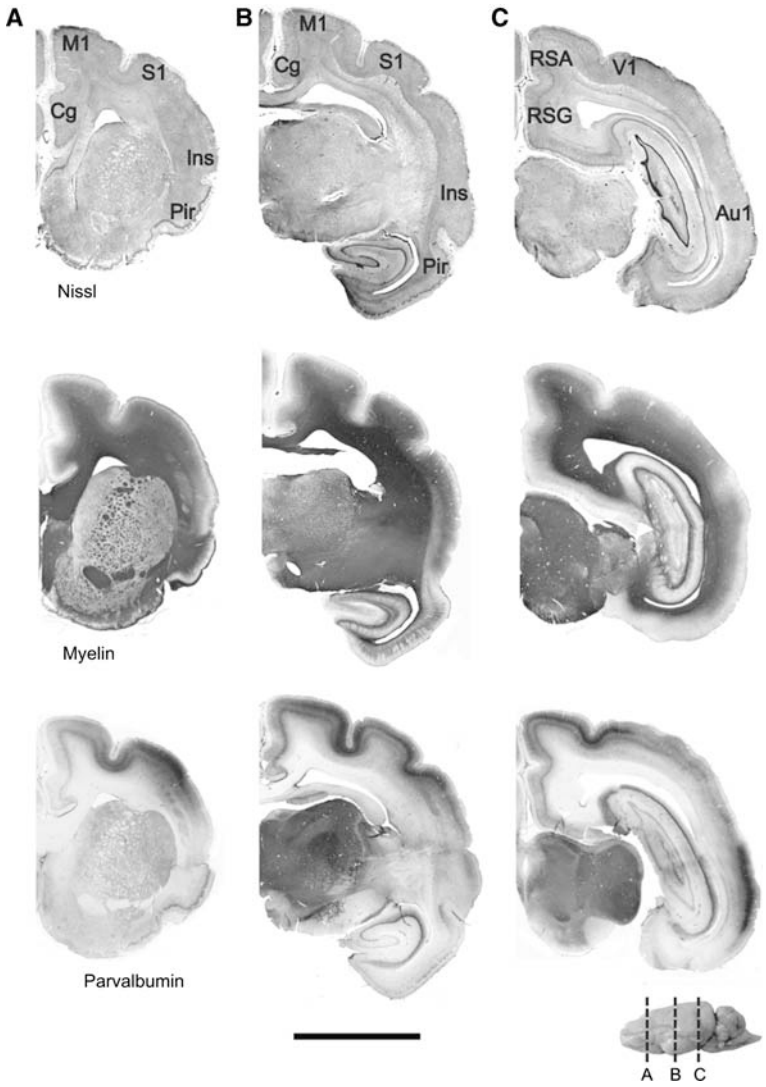


Fig. 3 Examples of coronal sections through rostral to caudal levels (**a**–**c**) of the lesser anteater brain, showing staining for Nissl substance, myelin, and PV. Note the clear definition of primary sensory cortices by intense myelination and PV immunoreactivity. *Au1* primary auditory cortex, *Cg* cingulate cortex, *Ins* insular cortex, *M1* primary motor cortex, *Pir* piriform cortex, *RSA* agranular retrosplenial cortex, *RSG* granular retrosplenial cortex, *S1* primary somatosensory cortex, *V1* primary visual cortex. *Scale bar* 10 mm

generated using Neurolucida software with a 63× objective lens (Zeiss Plan-Apochromat, N.A. 1.4) and manually tracing all immunopositive cell somata and the full extent of their processes visible throughout the thickness of the section.

Quantification

To characterize further the chemoarchitecture of neocortex in these species, we quantified numerical densities of Nissl-stained neurons, CB-, CR-, PV-, and NPY-immunoreactive (-ir) interneurons within each layer of the primary somatosensory cortex. In addition, we estimated the density of NPNFP-ir neurons in layer V of anterior cingulate cortex, primary somatosensory cortex, and primary visual cortex. Estimates of neuronal densities were obtained using the optical disector with a fractionator sampling implemented in two sections for each cell type. Disector frames were set at 50 × 50 μm and grid stepping varied depending on the size of the reference area. On average, 161.2 ± 29.5 (mean ± standard deviation) counting frames per layer for each cell type were investigated. Disector analysis was performed under Koehler illumination using a 63× objective lens (Zeiss Plan-Apochromat, N.A. 1.4). The thickness of optical disectors was set to 6 μm to allow for a sufficient guard zone and optical measurement of mounted section thickness was collected at every eighth sampling site. Cellular densities were derived from these stereological counts and corrected for *z*-axis shrinkage from histological processing by the number-weighted mean measured section thickness as described previously (Sherwood et al. 2007).

Systematic random sampling of neuron volumes were also obtained as they were encountered during optical disector analysis for CB-, CR-, PV-, NPY-, and NPNFP-ir neurons by using the nucleator (Gundersen 1988) with a vertical probe axis set perpendicular to the pial surface and using four sampling rays.

Evolutionary reconstruction

The tree topology of mammals from Murphy et al. (2007) was used to map character state evolution. This phylogeny incorporates information from large nuclear and mitochondrial gene datasets, in addition to analysis of coding indels and retroposon insertions to further resolve the root of the placental mammals. The evolution of neocortical traits was optimized to the tree using maximum parsimony analysis with character state transformations unordered, as implemented in Mesquite software version 1.06 (Maddison and Maddison 2005). The maximum parsimony method of reconstructing character evolution minimizes the number of character state changes on the tree. We also mapped character state evolution using maximum likelihood Markov *k*-state model 1, which does not assume any state to be plesiomorphic at the root of the tree and which allows character states to change into others on any branch with equal probability. A decision threshold of 2.0 was used for statistical consideration. Branch lengths were set according to the method of Pagel (1992). Characters were coded based on direct observation for the specimens in the current study, supplemented by written description in the literature for other species (Ashwell et al. 2005; Boire et al. 2005; Desgent et al. 2005; Glezer et al. 1993, 1998; Hassiotis et al. 2005; Hof et al. 1996a, 1999; Van der Gucht et al. 2001) and examination of our own collections which include various marsupials, primates, rodents, carnivores, and cetartiodactyls. Because the marsupials represent a critical outgroup in defining the polarity of character evolution among eutherians, it should be noted that our determination of character states in Marsupialia for NPNFP-ir neurons was based on relatively limited available data from descriptions of the macropodid tammar wallaby (*Macropus eugenii*) in Ashwell et al. (2005), corroborated by our observations of NPNFP-stained sections from a parma wallaby (*Macropus parma*) in our own collection. To simplify the visualization of phylogenetic trees, whenever the same character state was consistently present in all members of a clade, the higher-order taxonomic designation is shown in the reconstruction of character state evolution.

Results

Nonphosphorylated neurofilament protein-containing neurons

The majority of NPNFP-ir neurons throughout the neocortex of these species were located in a band concentrated in layer V. In several regions, however, NPNFP-ir neurons were also sparsely distributed in layers III and VI. Across all species, the ventral portion of the anterior cingulate cortex, prelimbic cortex, infralimbic cortex, and orbital cortex contained relatively few NPNFP-ir neurons as compared to other regions. In contrast, darkly stained and dense NPNFP-ir neurons in layer V were observed in primary motor cortex, retrosplenial cortex, and primary visual cortex. In primary somatosensory and auditory cortex, layer III exhibited more prominent staining of NPNFP-ir neurons than elsewhere, showing mostly round and multipolar morphologies, with few pyramidal types. Within the insular cortex, the conspicuous band of staining present in layer V of other cortical areas was replaced by a more diffuse arrangement of NPNFP-ir somata spanning layers II–VI. Figure 4 shows the morphology and distribution of NPNFP-ir neurons in primary somatosensory cortex.

Although the patterns described above were generally similar across species, phylogenetic variation was evident. Notably, two-toed sloths displayed a relatively low density of layer V NPNFP-ir neurons in medial frontal cortex and retrosplenial cortex. In the giant anteater, there was a high concentration of NPNFP-ir neurons in layer III throughout the cortex compared to the other species.

Figure 5a illustrates the considerable variation among species in the proportion of the total Nissl-stained neuronal population that was immunostained for NPNFP in layer V of primary somatosensory cortex. The species mean proportion of NPNFP-ir neurons displayed a correlation with brain mass ($r_s = 1.0$, $P < 0.01$, $n = 4$ with two-toed sloth excluded); however, two-toed sloths showed a striking departure from this trend in having a substantially lower percentage of NPNFP-ir neurons than expected for their brain size (Fig. 5b).

It is also notable that neocortical NPNFP-ir neurons exhibited a high degree of diversity in their morphology. Many NPNFP-ir neurons displayed a typical pyramidal shape, defined by a basal dendritic skirt and a prominent apical dendrite extending to suprajacent layers. However, a high frequency of other morphologies was also evident. Some NPNFP-ir neurons included inverted pyramidal cells (Fig. 6h, l) and others displayed bifurcated apical dendrites that spread widely as they emerged from the perikaryon, resembling the "extraverted neurons" originally described in hedgehog, elephant shrew, and big brown bat by Sanides and Sanides (1974) (Fig. 6a, i). In addition to these, other variant multipolar (Fig. 6a, j, k) and fusiform (Fig. 6f, g) morphologies were observed. Among these fusiform types were club- or mace-shaped somata. Figure 5c shows the relative frequencies of these different classes of NPNFP-ir neurons in layer V of anterior cingulate cortex, primary somatosensory cortex, and primary visual cortex. Although atypical non-pyramidal dendritic arborization patterns were seen in many

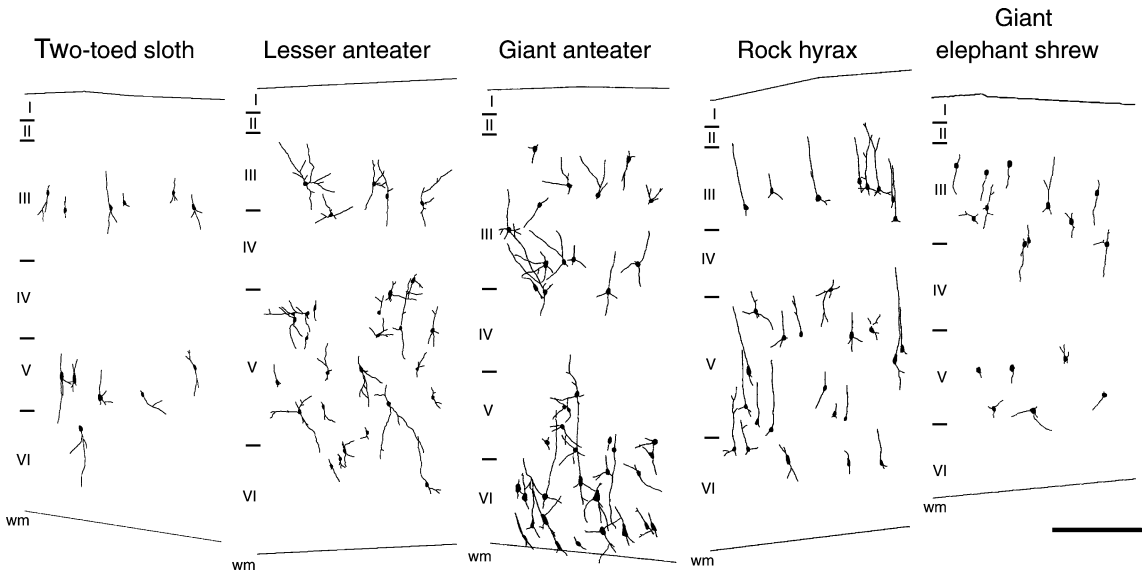


Fig. 4 Line drawings of NPNFP-ir neurons in the primary somatosensory cortex. *Roman numerals* indicate layers. *wm* White matter. *Scale bar* 500 μm

neurons, which might suggest disorganization of radial columnar architecture, it is important to note that vertical bundles of NPNFP-ir apical dendrites and myelinated axons were also observed throughout the cerebral cortex in all species (Fig. 6b).

To characterize further the differences between typical pyramidal NPNFP-ir neurons and the other non-pyramidal morphologies, we measured perikarya volumes of these cell classes in layer V of anterior cingulate cortex, primary somatosensory cortex, and primary visual cortex (Fig. 7). We used repeated-measures factorial ANOVA to determine whether there were significant differences in volume between pyramidal and non-pyramidal morphological types within each specimen, taking cortical area into account. Results demonstrated no significant effect of morphological type ($F_{1,522} = 2.95$, $P = 0.09$) or cortical area ($F_{2,522} = 1.63$, $P = 0.20$).

Neurons containing calcium-binding proteins and neuropeptide Y

In mammals, various calcium-binding proteins and neuropeptides are expressed in different subpopulations of GABAergic interneurons, providing a useful indication of the morphology and distribution of cells that participate in the intrinsic inhibitory interneuron system (Ascoli et al. 2008; Markram et al. 2004). In addition, calcium-binding proteins are expressed by subsets of projection neurons in particular phylogenetic groups (Hof et al. 1999; Hof and Sherwood 2005). Here we describe the staining patterns in the neocortex for CB, CR, PV, and NPY.

Calbindin

In all species, populations of darkly stained CB-ir cells were observed across layers II–VI. In layers II–III, CB-ir interneurons tended to have small- to medium-sized somata with round, bipolar, or bitufted morphologies (Fig. 8). In addition, layers IV–VI also contained medium-sized multipolar types with extensive dendritic ramifications. Figure 9 shows the distribution and morphology of CB-ir interneurons in primary somatosensory cortex. CB-ir interneurons occurred in low densities overall, with little regional variation throughout the neocortex. In contrast, the piriform cortex showed high concentrations of large multipolar CB-ir neurons in layer III. The neuropil contained diffuse CB-ir puncta and a relatively low density of vertically oriented smooth and varicose axon fiber segments. However, the organization of these axon fibers did not resemble the tightly interwoven double bouquet axon bundles that have been described in primates and carnivores (Ballesteros-Yañez et al. 2005).

Numerous lightly stained CB-ir cells were also observed in layers II–III of the neocortex with various pyramidal, extraverted, and multipolar shapes similar to NPNFP-containing neurons in adjacent sections (Fig. 8a, b). There was considerable regional variation in the density of these CB-ir presumptive projection neurons in layers II–III, with the highest frequencies found in frontal cortex, insular cortex, primary somatosensory cortex, primary motor cortex, and visual cortex. CB-ir pyramidal neurons have been similarly reported in layers II–III of rodents (Andressen et al. 1993; Desgent et al. 2005; Gonchar and Burkhalter 1997; Van Brederode et al. 1991), primates (Hof and Morrison 1991;

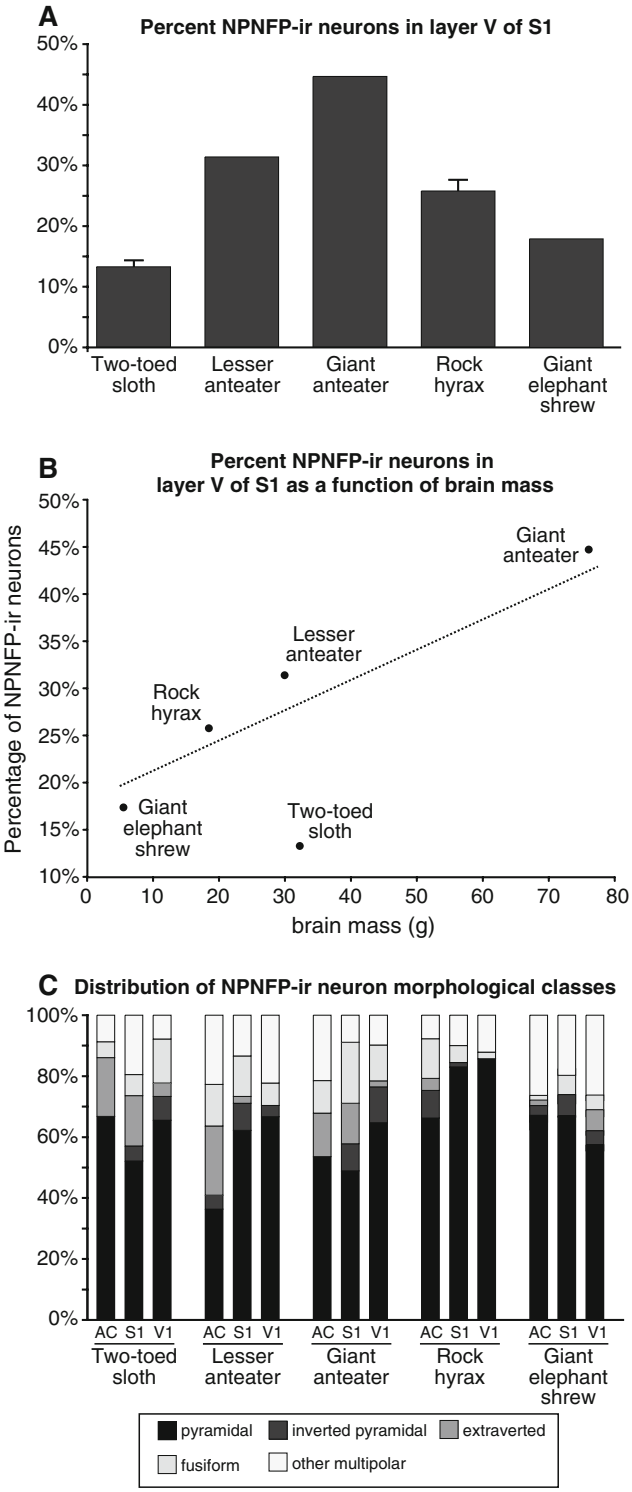


Fig. 5 Quantification of NPNFP-ir neuron numbers and morphological types. **a** *Bar graph* of the mean percentage of the total Nissl-stained neuron population in layer V of primary somatosensory cortex (*S1*) that was NPNFP immunoreactive. When more than one individual within a species was examined, the *error bar* represents the standard deviation. **b** Species mean percentage of NPNFP-ir neurons in layer V of the primary somatosensory cortex plotted against species mean brain mass. A least-squares regression line is fit to all data excluding the two-toed sloth. **c** The relative frequency of different morphological classes of NPNFP-ir neurons in layer V of the anterior cingulate cortex (*AC*), the primary somatosensory cortex (*S1*), and the primary visual cortex (*V1*). When more than one individual within a species was examined, the mean is shown

Kondo et al. 1999; Sherwood et al. 2004, 2007), megachiropterans, perissodactyls, and cetartiodactyls (Hof et al. 1999). In addition, CB-ir atypical pyramidal neurons have been described in layer IIIc/V of the visual cortex of cetaceans, including inverted pyramidal neurons (Glezer et al. 1993).

Calretinin

The neuron types expressing CR were generally comparable in all the species examined. Immunostaining against CR revealed very sparse, darkly labeled, small bipolar and tripolar cells with ovoid or round somata (Fig. 10). CR-ir interneurons were distributed throughout layers II–VI of the neocortex (Fig. 11). While layers II–III contained mostly small bipolar neurons, occasional medium multipolar cells were also seen intermingled among bipolar cells in layers V–VI. Additionally, a number of CR-ir neurons were observed in layer I, putatively corresponding to Cajal-Retzius cells. These cells had round or ovoid somata with proximal dendrites oriented in an axis parallel to the pia. Cajal-Retzius cells that express CR are known from many other mammals (Cruikshank et al. 2001; Glezer et al. 1998; Gonchar and Burkhalter 1999; Hassiotis et al. 2005; Hof et al. 1999; Schwark and Li 2000; Sherwood et al. 2004). Although there was generally not very intense CR staining of the neuropil, the detectable immunoreactive fibers were beaded and usually had a vertical orientation (Fig. 10a). Overall, regional variation in the distribution of CR-ir interneurons in the neocortex was not pronounced. The densities of neocortical CR-ir interneurons were very low in all species with the exception of the giant elephant shrew, which displayed a relatively higher incidence of these cell types.

Parvalbumin

In all species examined, PV-ir cells were mostly multipolar neurons distributed in layer II–VI, with the highest concentration in layers III–V (Fig. 12). This laminar distribution is similar to all other mammals investigated to date (Alcantara and Ferrer 1994; Desgent et al. 2005; Dhar et al. 2001; Glezer et al. 1993; Goodchild and Martin 1998; Hassiotis et al. 2005; Hendry and Jones 1991; Hof et al. 1996b, 1999; Sherwood et al. 2007; Spatz et al. 1994). Many of the PV-ir neurons had morphologies resembling the basket cells that have been described in other mammals (Hof et al. 1999). Large PV-ir multipolar cells typically had

a horizontally spread dendritic tree that could be followed for a distance from the parent soma (Fig. 13). However, PV-ir neurons in supragranular layers tended to have smaller somata sizes and more vertically oriented dendritic arbors than those in deeper layers.

There was regional variation in the laminar patterns, overall density of immunoreactive somata, and neuropil staining for PV. In all species, there was a striking low density of PV staining in medial frontal cortex, cingulate cortex, motor cortex, piriform cortex, and insular cortex. This contrasts with the high density of PV-ir cells and neuropil staining in layer IV of heavily-myelinated regions corresponding to primary somatosensory cortex, auditory cortex, and visual cortex.

The hyrax, sloth, and lesser anteater specimens showed particularly distinct modular patches of PV-ir neuropil staining in layer IV of primary somatosensory and auditory cortex (Fig. 12a). Similar patches of PV neuropil staining, which correspond to the focused terminations of thalamocortical afferents in layer IV, have also been noted in sensory areas of primates and rodents (DeFelipe and Jones 1991). In these same regions, as well as visual cortex, vertical arrays of PV-ir boutons could be observed in layers II–IV (Fig. 12b). Electron microscopy has demonstrated that these PV-ir cartridges correspond to the terminations of chandelier cells as they synapse on the axon initial segment of pyramidal neurons (Fairén and Valverde 1980; Peters et al. 1982).

Neuropeptide Y

All species showed a comparable pattern of immunostaining to NPY. Neurons that stained for NPY were mostly medium multipolar types, but also included some medium-sized round and bipolar cells (Fig. 14). In general, the distribution of NPY-ir somata across the neocortex was very sparse, with the highest densities in layers V–VI (Fig. 15). NPY-ir neurons were also located in the underlying white matter. In all species, many lightly stained NPY-ir pyramidal neurons were seen, with the highest concentration in layer V. A plexus of NPY-ir axon fibers could be observed throughout the neuropil of the cortex. There was no clear regional variation in the staining pattern for NPY across the neocortex.

Quantitative analysis of interneurons

Estimates of numerical densities for different classes of neurochemically-defined interneurons in the primary somatosensory cortex of the five species are shown in Fig. 16. For these quantitative estimates, we counted only the most intensely immunoreactive cells with morphologies putatively corresponding to interneuron classes. Among these species, layers II and III generally contained the highest density of CB-ir and CR-ir interneurons, whereas PV-ir interneurons were relatively denser in layers IV–VI. Interneurons immunoreactive for NPY showed distributions comparable to the calcium-binding proteins, with a tendency to increase in density within deeper cortical layers.

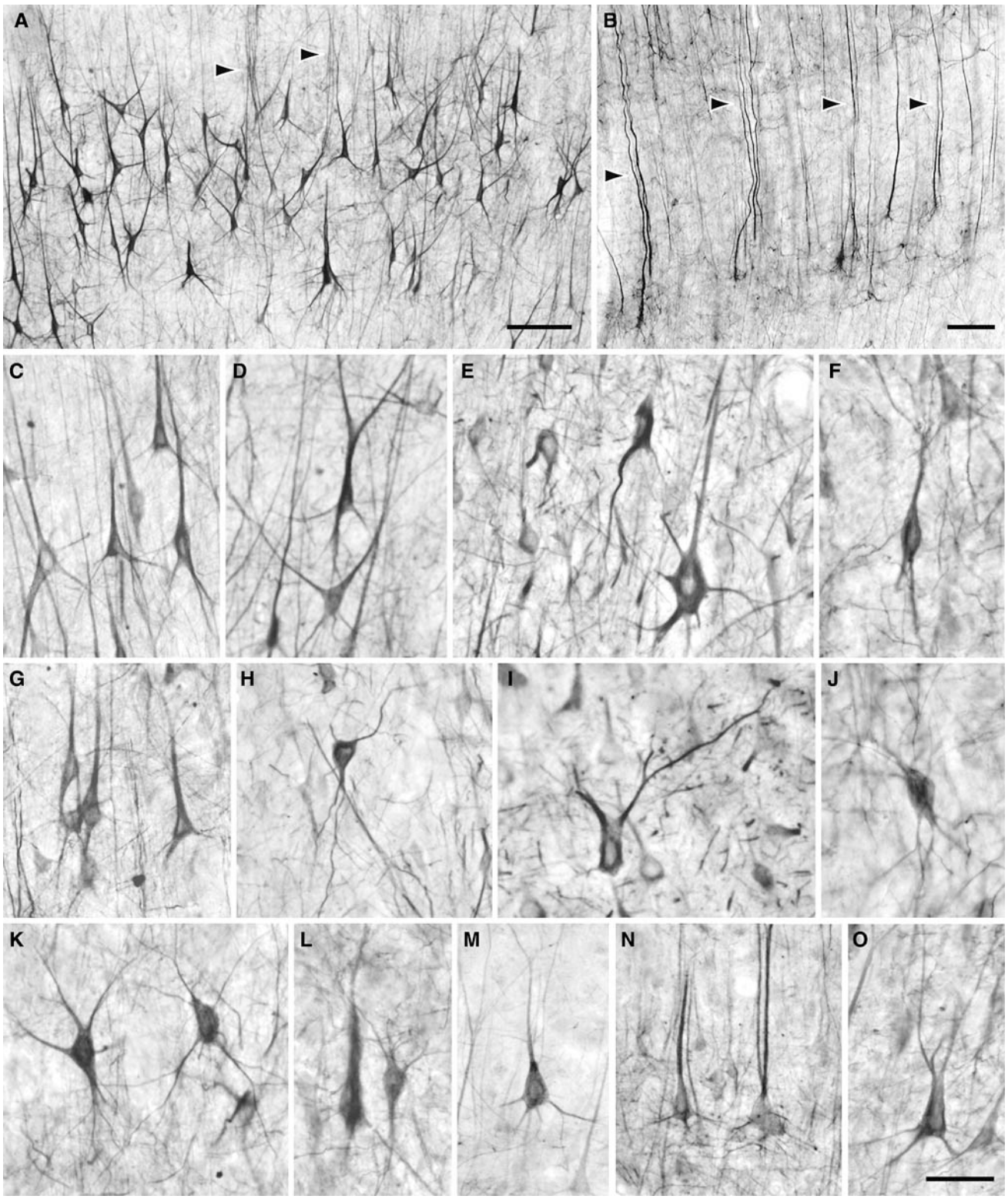
Figure 17 displays the mean cellular volume of different interneuron classes from the primary somatosensory cortex. Repeated-measures ANOVA of within-subjects variation in cellular volumes revealed a significant effect of interneuron type ($F_{3,674} = 53.62$, $P < 0.0001$). Bonferonni post-hoc contrasts among interneurons types are shown in Table 2. The PV-ir neuron soma volumes were, on average, larger than all other neurochemically-defined interneuron types within individuals. Similarly, neocortical PV-ir neurons have also been found to be larger than CB- and CR-ir neurons in primates and cetaceans (Bourne et al. 2007; Dhar et al. 2001; Glezer et al. 1993, 1998; Sherwood et al. 2007; Van Brederode et al. 1990). In our sample, the somatic volume of NPY-ir neurons was also usually among the largest, which is consistent with the characterization of these cells as basket types (Toledo-Rodriguez et al. 2005).

Glial fibrillary acidic protein

In all regions of the cortex among all species examined, staining against GFAP revealed a diffuse network of stellate astroglial cells with very short, branching, primary processes arising from the cell soma (Fig. 18). The greatest GFAP immunoreactivity was found in layers I–III. This pattern of GFAP-ir astroglial architecture is similar to that found across most mammals described previously, including marsupials, artiodactyls, carnivores, rodents, scandentians, and chiropterans (Colombo et al. 2000). This contrasts with the GFAP-ir astroglial organization of the primate cerebral cortex, which is predominantly characterized by long, unbranched, radially-oriented interlaminar processes spanning supragranular cortical layers (Colombo et al. 1998, 2004).

Phenetic affinities of cortical chemoarchitecture

We used multivariate techniques to explore whether there is congruence between phylogenetic relationships and the chemoarchitecture of the primary somatosensory cortex. For these analyses, we calculated the percentage of the total Nissl-stained neuron population within layers II–VI that were immunostained for CB, CR, PV, and NPY. In addition, we calculated the percentage of the Nissl-stained neuron population in layer V that were immunostained for NPNFP and the proportion of the NPNFP-ir neurons that were non-pyramidal in morphology. Data used in these multivariate analyses are shown in Table 3. We used



principal components analysis of the covariance matrix derived from these six measures to summarize variation among specimens.

The first two principal components (PC) captured 96.6% of the variance in these data (Fig. 19a). Principal component 1 (PC1) explained 59.0% of the variance and showed

◀ **Fig. 6** Photomicrographs of NPNFP immunostaining in the neocortex of various atlantogenatan species. **a** Layer V of the dorsal frontal cortex in a giant anteater, showing a variety of neuron morphological types. The *arrowheads* indicate apical dendrite bundles. **b** Layer V of the primary visual cortex in a rock hyrax, showing clusters of neurons and bundling of their apical dendrites. The *arrowheads* indicate apical dendrite bundles. **c** Layer V of the dorsal frontal cortex in a giant anteater. **d** Layer V of the primary somatosensory cortex in a two-toed sloth. **e** Layer V of the primary motor cortex in a giant anteater. **f** Layer V of the primary somatosensory cortex in a lesser anteater, showing a fusiform neuron. **g** Layer V of the primary visual cortex in a rock hyrax. **h** Layer V of the primary somatosensory cortex in a giant anteater, showing an inverted pyramidal neuron. **i** Layer V of the anterior cingulate cortex in a lesser anteater, showing an extraverted neuron. **j** Layer V of the dorsal frontal cortex in a giant anteater. **k** Layer III of the primary auditory cortex in a two-toed sloth, showing multipolar neurons. **l** Layer V of the primary somatosensory cortex in a lesser anteater, showing a typical pyramidal neuron alongside an inverted pyramidal neuron. **m** Layer V of the primary motor cortex in a rock hyrax. **n** Layer V of the primary visual cortex in a two-toed sloth. **o** Layer V of the anterior cingulate cortex in a lesser anteater. *Scale bar* in **a**–**b** 50 μm. *Scale bar* in **c**–**o** 20 μm

the greatest loading on the percentage of layer V neurons that were NPNFP-ir (Table 4). The factor coordinates on PC1 showed pronounced overlap among afrotherians and xenarthrans, indicating that this factor did not clearly ordinate taxa according to phylogeny. PC2 explained 37.6% of the variance and loaded most on the percentage of non-pyramidal NPNFP-ir neurons and the proportion of NPY-ir interneurons. This axis was more effective at separating xenarthrans and afrotherians. To examine whether the PC factor coordinates of cases were related to overall scaling effects, we calculated their correlation with brain mass. Neither PC1 ($r = 0.64$, $P = 0.12$, $n = 7$) nor PC2 ($r = 0.18$, $P = 0.70$, $n = 7$) were correlated with brain mass.

Next, we calculated a phenogram with the unweighted pair-group method using arithmetic average (UPGMA) based on Euclidean D^2 distances to examine whether hierarchical clustering of these multivariate data mirrors known phylogenetic affinities. Figure 19b shows that phenetic similarities based on this analysis did not concord with phylogenetic relationships, although it is notable that individuals from the same species were grouped together.

Character state evolution

We used maximum parsimony and maximum likelihood techniques to reconstruct the evolution of semi-quantitative characters based on the current observations, in conjunction with descriptions from the literature (Fig. 20). Results showed that some neocortical character states represent synapomorphies that exclusively define

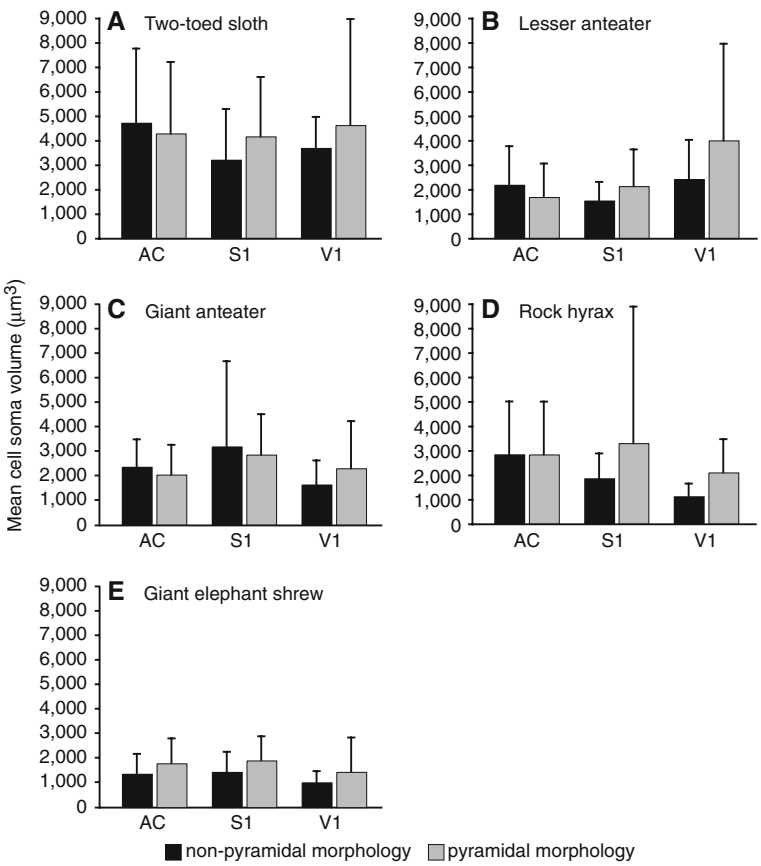


Fig. 7 *Bar graphs* showing the soma volumes of NPNFP-ir subtypes in the anterior cingulate cortex (*AC*), the primary somatosensory cortex (*S1*), and the primary visual cortex (*V1*). The different varieties of non-pyramidal neuron morphologies (inverted pyramidal, extraverted, fusiform, and other multipolar) were combined in this analysis. Because initial analyses did not reveal a significant difference in cell type-specific soma volumes between individuals of the same species, data were pooled in cases where more than one individual was available in a species. Note that these volumes are not corrected for tissue shrinkage due to fixation and histological processing; however, the volumes of different subtypes within the same individual are equally affected by shrinkage. *Error bars* represent standard deviation

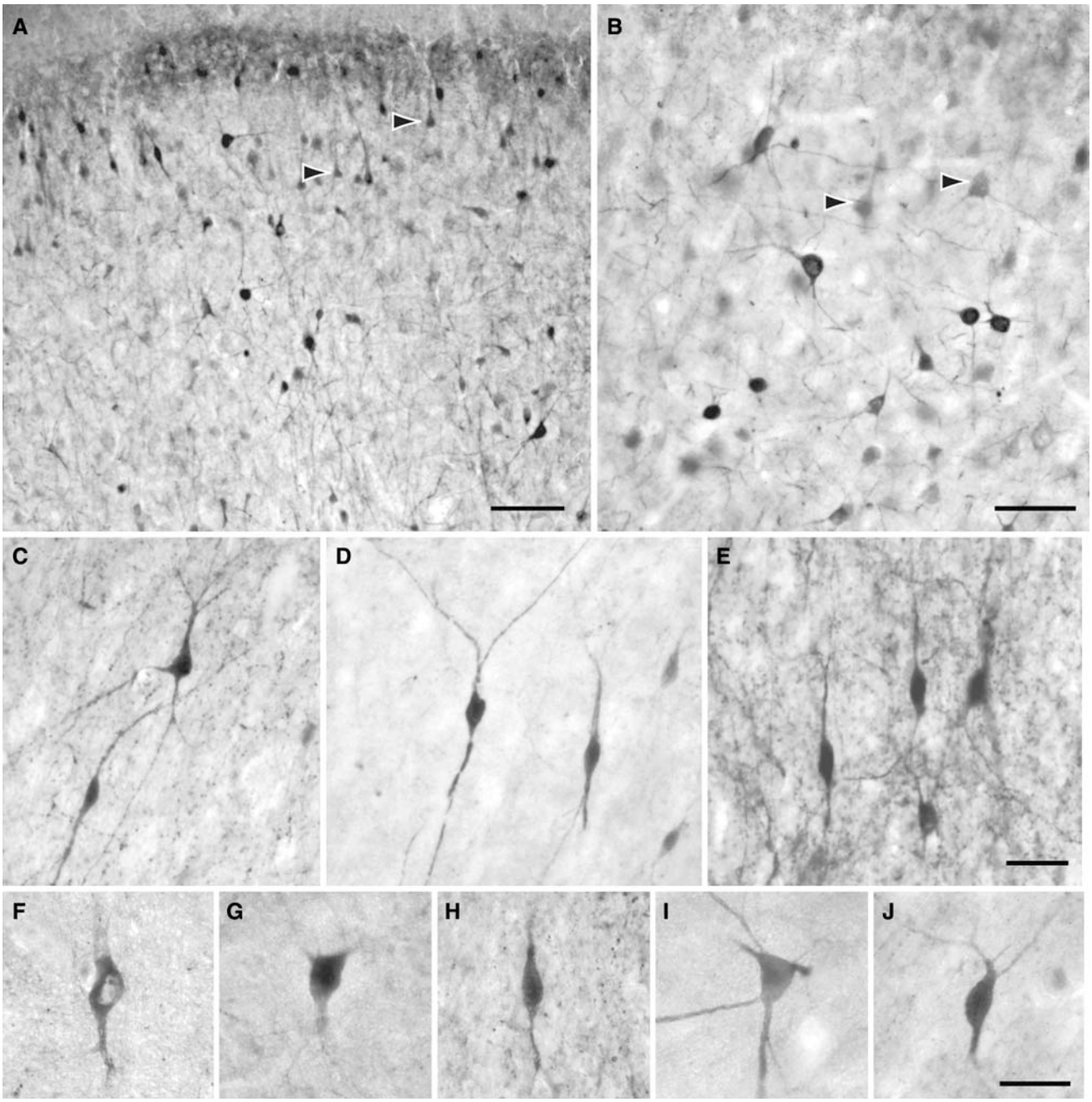


Fig. 8 Photomicrographs of CB immunostaining in the neocortex of various atlantogenatan species. **a** Layers I–III of the primary somatosensory cortex in a giant elephant shrew. **b** Layer III of the anterior cingulate cortex in a rock hyrax. *Arrowheads* in **a** and **b** indicate CB-ir pyramidal neurons. **c** Layer III of the primary visual cortex in a lesser anteater. **d** Layer II of the primary auditory cortex in a lesser anteater. **e** Layer III of the primary motor cortex in a two-toed sloth. **f**, **g** Layer V of the primary somatosensory cortex in a rock hyrax. **h**, **i** Layer III of the anterior cingulate cortex in a lesser anteater. **j** Layer VI of the primary visual cortex in a giant anteater. *Scale bars* in **a** 100 μm, **b** 80 μm, **c**–**e** 50 μm, **f**–**j** 20 μm

particular clades of species, whereas other characters display a striking degree of homoplasy. Specifically, the relative frequency of different calcium-binding protein containing interneuron types appeared to be especially prone to convergent evolution. For example, marsupials, on the one hand, and cetartiodactyls and perissodactyls, on the other hand, are similar in having a paucity of PV-ir interneurons; however, this trait is not inherited by descent because these taxa do not share a recent common ancestor. Furthermore, a low density of CR-ir interneurons has evolved independently in megachiropterans to resemble the condition in monotremes and most

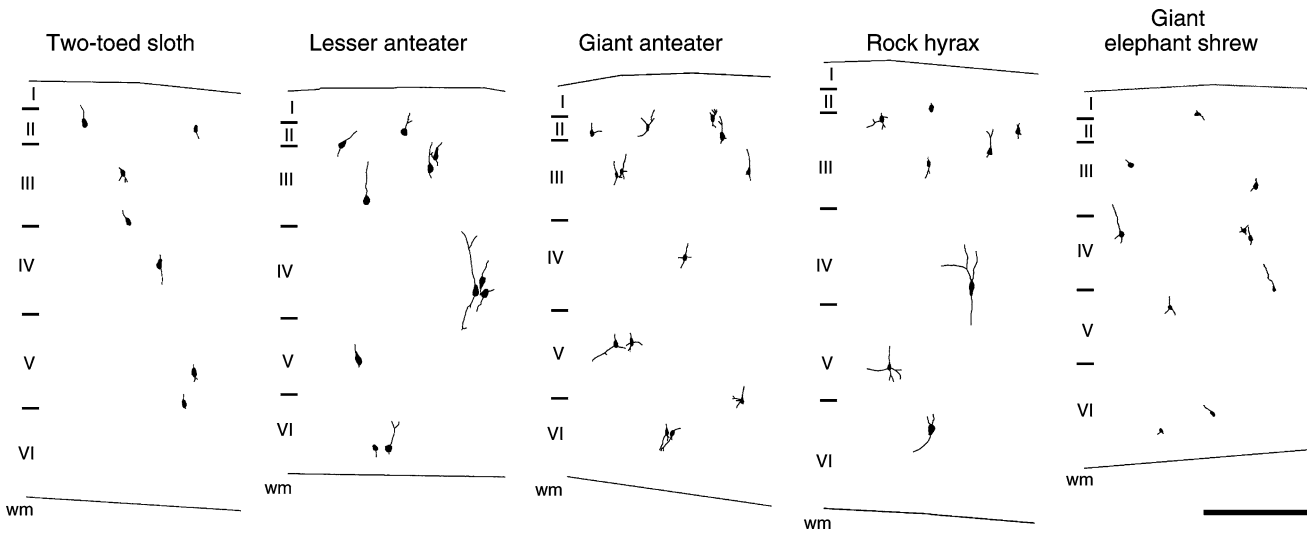


Fig. 9 Line drawings of CB-ir interneurons in the primary somatosensory cortex. *Roman numerals* indicate layers. *wm* White matter. *Scale bar* 500 μm

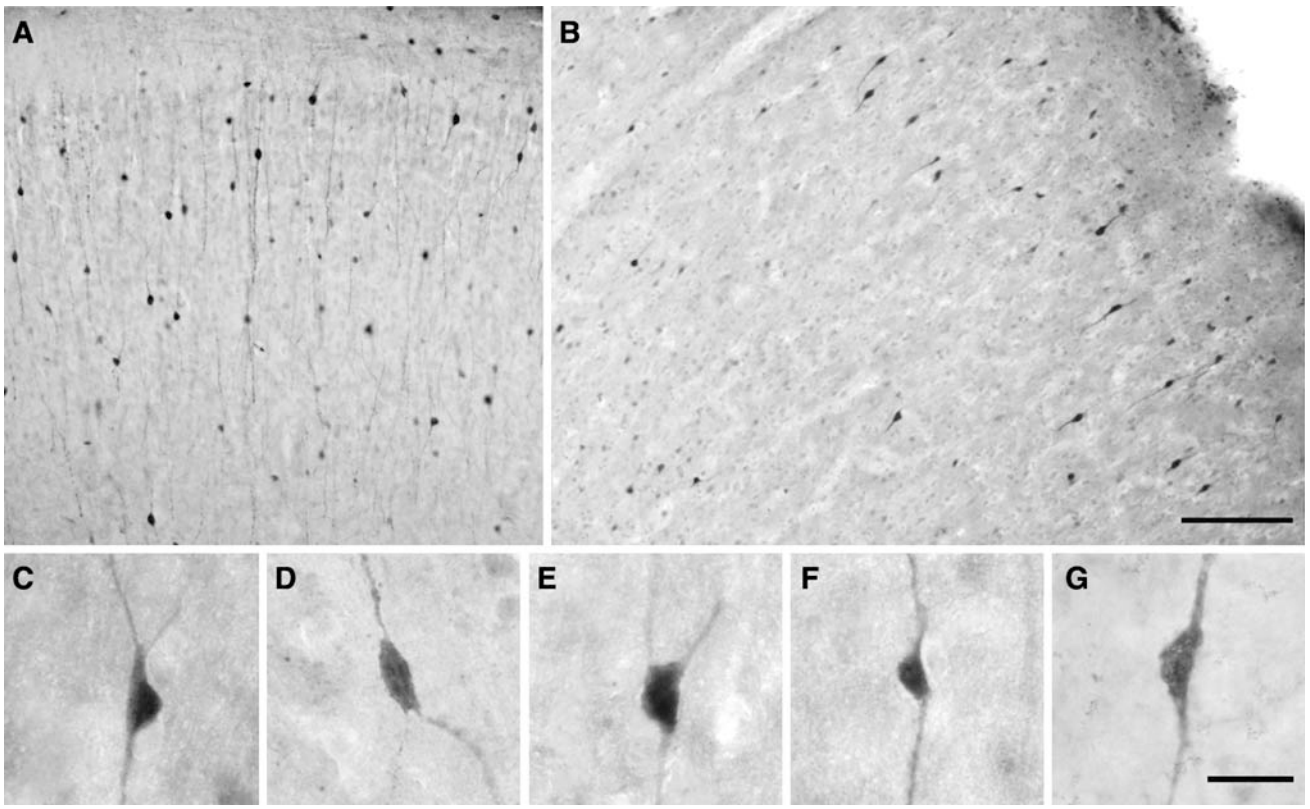


Fig. 10 Photomicrographs of CR immunostaining in the neocortex of various atlantogenatan species. **a** Layers I–III of the primary somatosensory cortex of a giant elephant shrew, showing many vertically aligned CR-ir axons. **b** Layers I–IV of the primary visual cortex in a rock hyrax. **c** Layer III of the primary motor cortex in a tow-toed sloth. **d**, **e** Layer III of the insular cortex in a rock hyrax. **f** Layer V of the primary motor cortex in a lesser anteater. **g** Layer III of the insular cortex in a giant anteater. *Scale bar* in **a**, **b** 100 μm, **c**, **g** 20 μm

atlantogenatans. The independent evolution of similar phenotypes in separate lineages echoes recent evidence that a specialized class of spindle-shaped projection neuron in the anterior cingulate and insular cortex, called the von Economo neuron, is distributed among several disparate clades, including hominids (i.e., great apes and

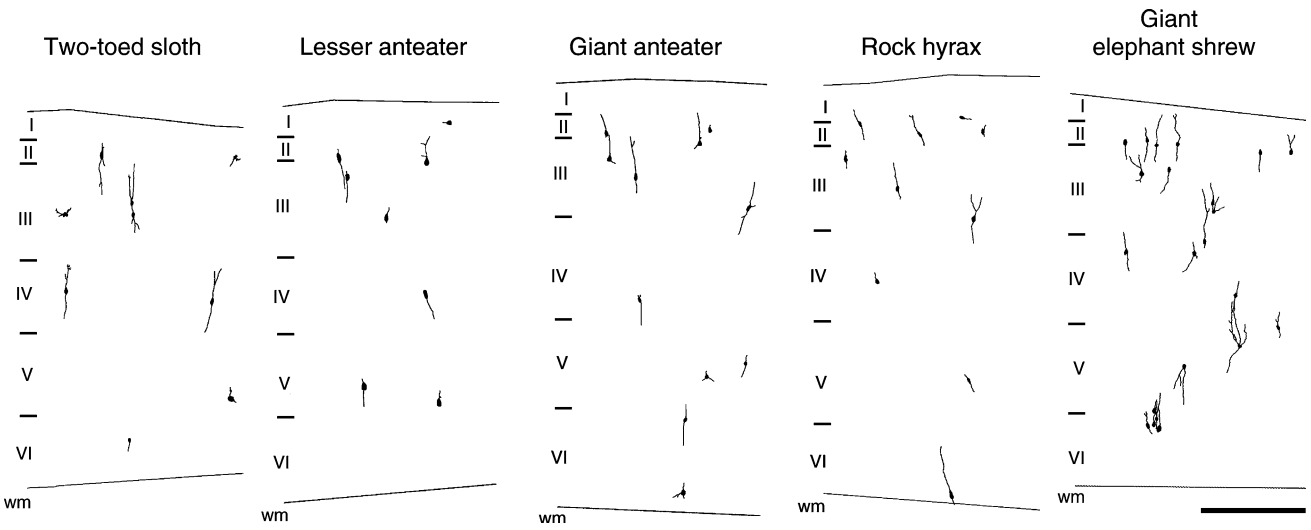


Fig. 11 Line drawings of CR-ir interneurons in the primary somatosensory cortex. *Roman numerals* indicate layers. *wm* White matter. *Scale bar* 500 μm

humans), cetaceans, and elephants (Hakeem et al. 2008; Hof and Van der Gucht 2007; Nimchinsky et al. 1999).

Putting the current data into a phylogenetic framework allowed us to reconstruct the neocortical organization of ancestral mammalian species (Table 5). The most parsimonious reconstruction of the stem mammal shows a distribution of neocortical neuron types that broadly resembles the condition of living monotremes and marsupials. The ancestral character state reconstruction for Eutheria, which is also supported by likelihood estimates, furthermore, favors the conclusion that xenarthran and afrotherian descendents have retained many conservative traits, whereas the boreoeutherian clade (Laurasiatheria + Euarchontoglires) is more derived in the overall distribution of neocortical cell types.

Discussion

Understanding the distribution of neocortical cell types in xenarthrans and afrotherians reveals an enriched view of the diversity of brain organization in mammals. Furthermore, because the xenarthan and afrotherian clades are joined in the Atlantogenata branch at the root of the eutherian tree, our findings help to reconstruct the sequence in which neocortical characteristics have evolved among mammalian taxa. Overall, atlantogenatan species most closely resemble monotremes and marsupials in the morphological diversity and laminar distribution of NPNFP-ir neurons, the proportions of calcium-binding protein containing interneurons, and the typology of GFAP-ir astrocytes. Therefore, our current findings in selected species of Xenarthra and Afrotheria support the view that the Atlantogenata retained a greater number of plesiomorphic traits during evolution compared to the more derived character states seen in members of the Boreoeutheria.

Morphology and distribution of projection neurons

This study presents the first description of immunoreactivity to NPNFP using the SMI-32 antibody in xenarthran and afrotherian neocortex. The laminar pattern of staining in atlantogenatans most closely resembles the monotreme echidna (Hassiotis et al. 2005) and the marsupial tammar wallaby (Ashwell et al. 2005), but differs from the distribution pattern of NPNFP-ir neurons in layer III, V, and VI throughout the neocortex which characterizes most other eutherians (Baldauf 2005; Boire et al. 2005; Bourne et al. 2005, 2007; Budinger et al. 2000; Campbell and Morrison 1989; Chaudhuri et al. 1996; Hof et al. 1996a; Hof and Sherwood 2005; Nimchinsky et al. 1997; Preuss et al. 1997; Sherwood et al. 2004; Tsang et al. 2000; Van der Gucht et al. 2007, 2001).

Immunoreactivity to NPNFP has been shown to display pronounced regional variation in staining across layers and selectivity for specific efferent neuron types in mammalian neocortex. Because of this heterogeneity, NPNFP immunoreactivity has been used to guide anatomical parcellation of the neocortex in monotremes (Hassiotis et al. 2004), marsupials (Ashwell et al. 2005), primates (Chaudhuri et al. 1996; Hof and Morrison 1995; Preuss et al. 1997; Sherwood et al. 2003; Van der Gucht et al. 2006), rodents (Boire et al. 2005; Kirkcaldie et al. 2002; Paxinos et al. 1999; Van der Gucht et al. 2007), and carnivores (Hof et al. 1996a; Van der Gucht et al. 2001). In the current study, immunostaining for NPNFP in the atlantogenatan species revealed regional variation that resembled the patterns described from these other mammals. Specifically, the

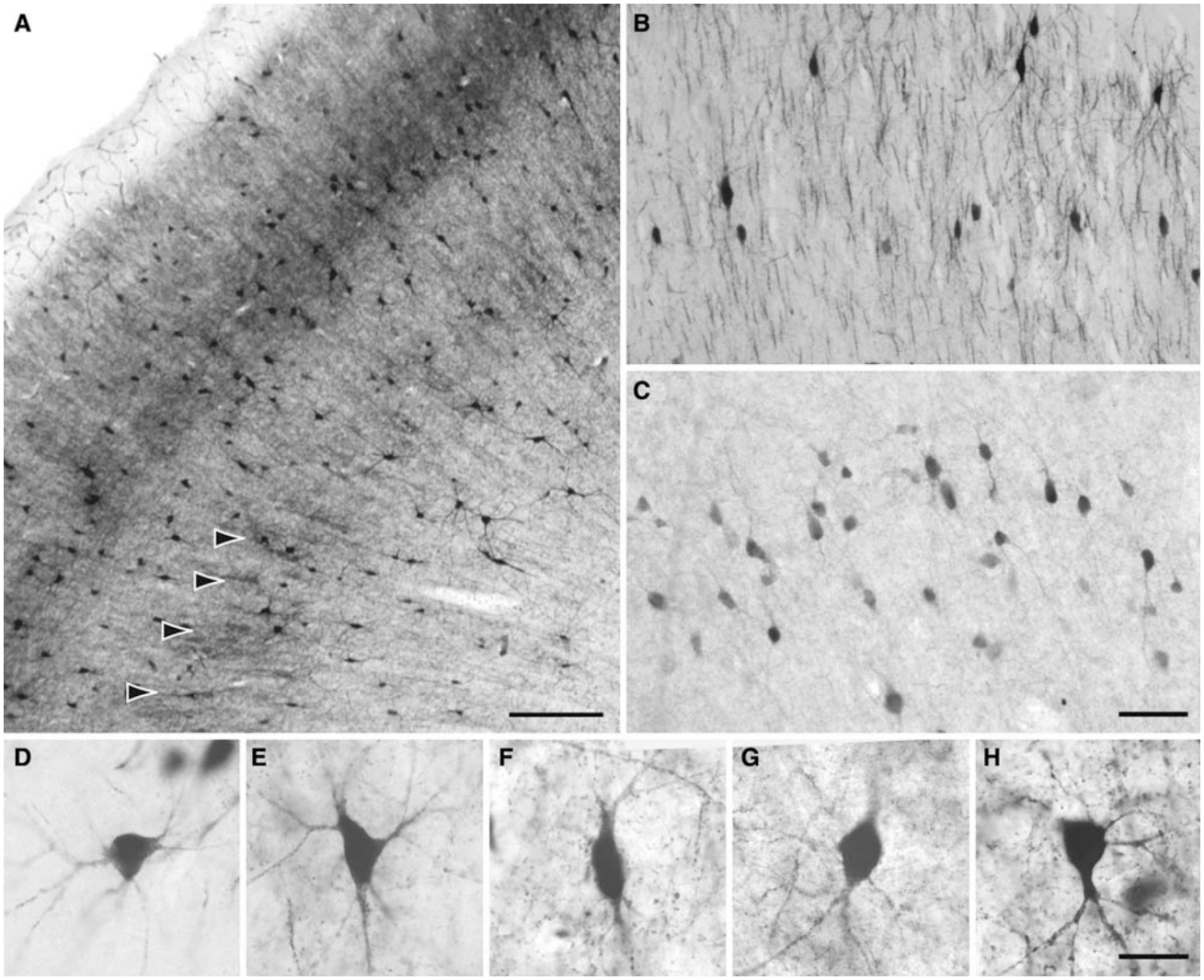


Fig. 12 Photomicrographs of PV immunostaining in the neocortex of various atlantogenatan species. **a** The primary auditory cortex of a lesser anteater. *Arrowheads* indicate the location of patches of neuropil staining in layer IV. **b** Layer III of the primary somatosensory cortex in a lesser anteater, showing PV-ir somata and cartridges. **c** Layer III of the primary somatosensory cortex in a rock hyrax. **d** Layer VI of the primary visual cortex in a giant elephant shrew. **e** Layer IV of the primary somatosensory cortex in a lesser anteater. **f** Layer III of the dorsal frontal cortex in a giant anteater. **g** Layer III of the anterior cingulate cortex in a two-toed sloth. **h** Layer V of primary somatosensory cortex in a rock hyrax. *Scale bar* in **a** 100 μm, **b**, **c** 50 μm, **d**–**h** 20 μm

prelimbic cortex, infralimbic cortex, and orbital cortex appeared to have relatively few NPNFP-ir neurons, whereas staining was most intense in primary somatosensory and auditory areas.

Among atlanotgenatans, we found a correlation between brain mass and the percentage of NPNFP-ir neurons in layer V. Because a similar relationship has been noted in primates (Sherwood and Hof 2007), it appears that NPNFP-ir neuron proportions show covariance with brain size as a product of scaling trends related to constraints on efferent connectivity. Neurofilament proteins are essential for the stabilization of the axon cytoskeleton (Morris and Lasek 1982) and their presence has been correlated with large axon diameter, rapid conduction velocity, and heavy myelination (Kirkcaldie et al. 2002; Lawson and Waddell 1991). In macaque monkeys, it has been demonstrated that the percentage of NPNFP-ir neurons increases in areas furnishing higher-order corticocortical and callosal connections (Campbell and Morrison 1989; Hof et al. 1995, 1996c). Furthermore, in rats it has been shown that NPNFP-immunoreactivity is most intense among neurons with specific intracortical projections, although it does not correlate with axon projection distance (Kirkcaldie et al. 2002). Thus, brain size enlargement might require higher levels of NPNFP expression within neurons to support the cytoskeleton for specific association connections.

It is remarkable, however, that the two-toed sloth demonstrated a considerably lower percentage of NPNFP-

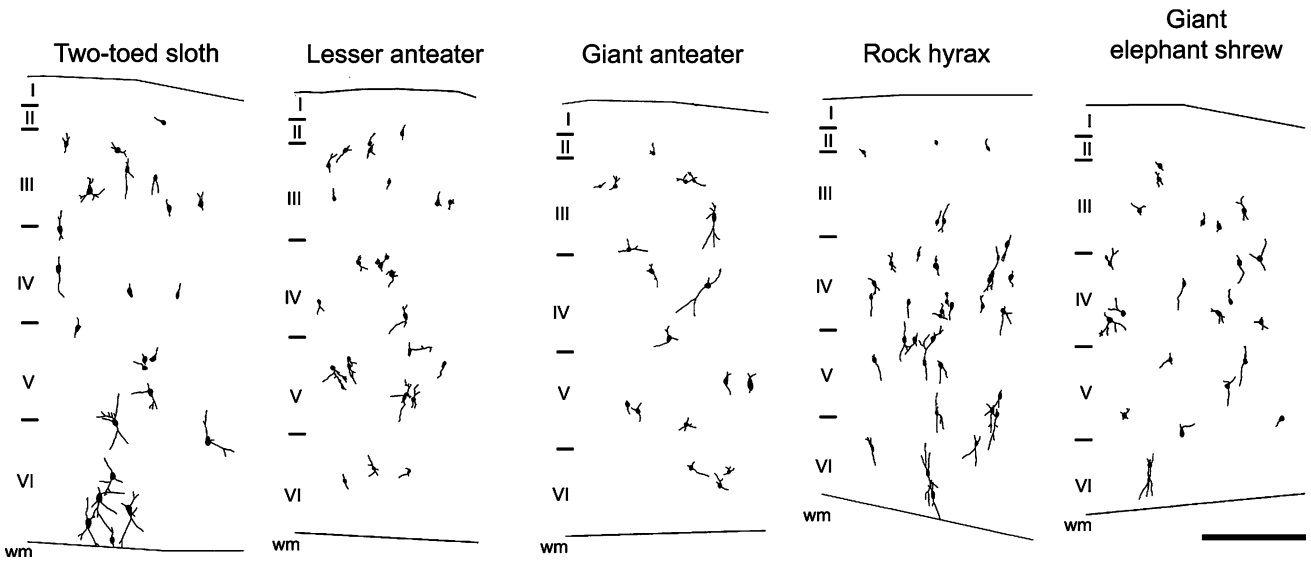


Fig. 13 Line drawings of PV-ir interneurons in the primary somatosensory cortex. *Roman numerals* indicate layers. *wm* White matter. *Scale bar* 500 μm

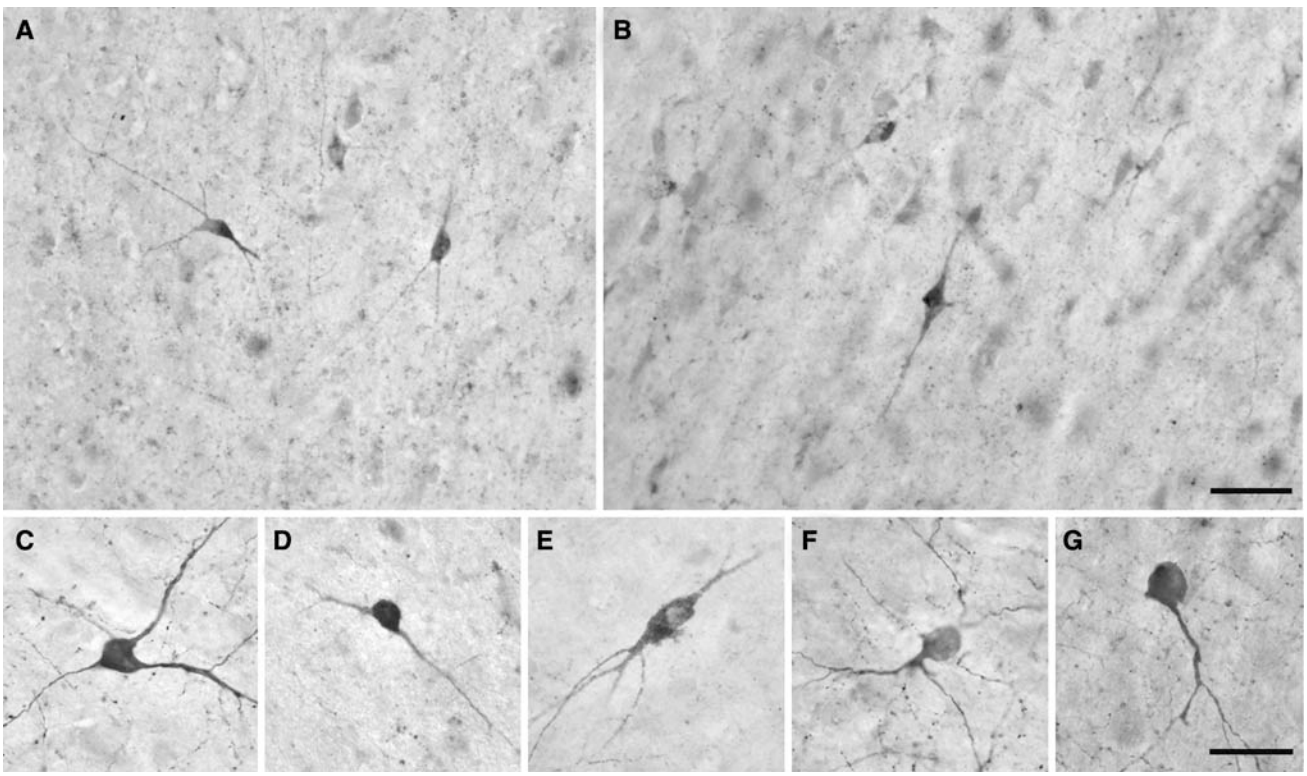


Fig. 14 Photomicrographs of NPY immunostaining in the neocortex of various atlantogenatan species. **a** Layer IV of the primary somatosensory cortex in a two-toed sloth. **b** Layer V of the insular cortex in a giant anteater. **c** Layer III of the primary visual cortex in a giant anteater. **d** Layer VI of the primary auditory cortex in a rock hyrax. **e** Layer V of the anterior cingulate cortex in a two-toed sloth. **f**, **g** Layer VI of the insular cortex in a rock hyrax. *Scale bar* in **a**, **b** 50 μm, **c**–**g** 20 μm

ir neurons in layer V than expected for its brain mass. Two-toed sloths are known for their extremely deliberate movements, slow digestion, reduced metabolic rates, and increased periods of sleep (Vizcaíno and Loughry 2008). Given the high metabolic costs of propagating action potentials along axons (Lennie 2003), it may be speculated

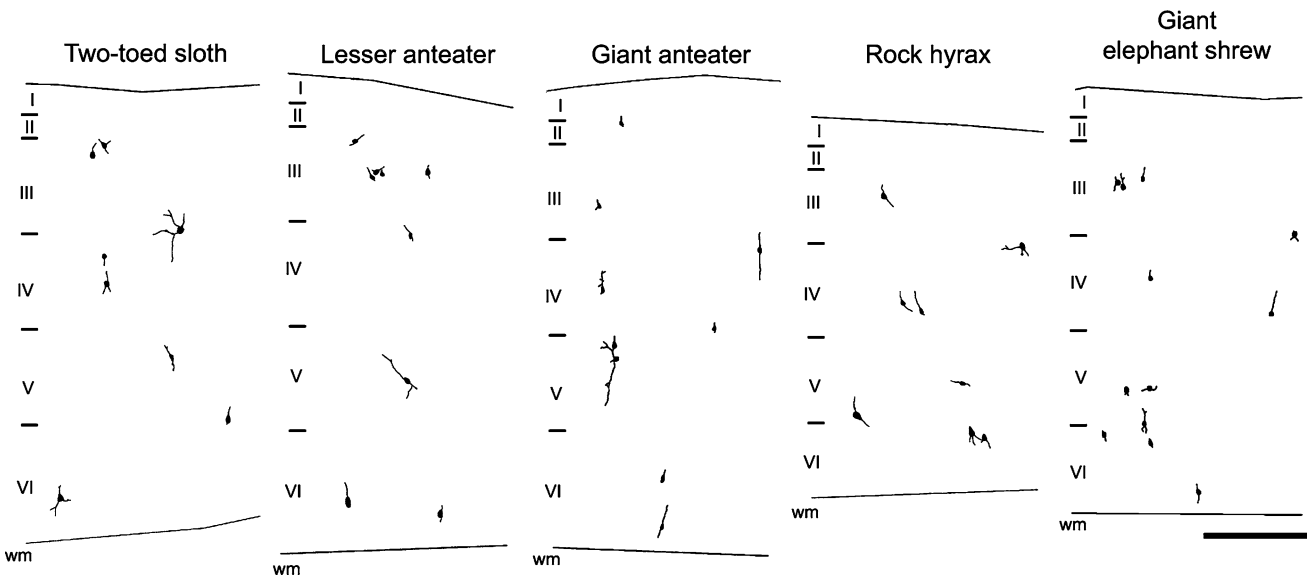


Fig. 15 Line drawings of NPY-ir interneurons in the primary somatosensory cortex. *Roman numerals* indicate layers. *wm* White matter. *Scale bar* 500 μm

that there has been compensatory energetic savings in the sloth brain by reducing the number of NPNFP-containing neurons with large diameter, heavily-myelinated axons.

Among all species of Xenarthra and Afrotheria, we observed several uncommon morphological classes of NPNFP-ir neurons alongside the more typical pyramidal cells. Similar non-pyramidal projection neurons have been described in the neocortex of other mammals. For example, neurons with extraverted morphologies have previously been noted in the quokka, hedgehog, elephant shrew, big brown bat, and humpback whale (Hof and Van der Gucht 2007; Sanides and Sanides 1974; Tyler et al. 1998). In addition, CB- and CR-ir extraverted neurons have been reported in several cetaceans (Glezer et al. 1998). Spiny and aspiny inverted pyramidal neurons have also previously been described in many other mammalian species, such as rodents, lagomorphs, primates, carnivores, and cetaceans (Garey et al. 1985; Glezer and Morgane 1990; Mendizabal-Zubiaga et al. 2007; Qi et al. 1999). Depending on the region and species, 1–8.5% of profiles represented inverted pyramidal neurons among rats, rabbits, chimpanzees, and cats (Bueno-Lopez et al. 1991; Mendizabal-Zubiaga et al. 2007; Parnavelas et al. 1977; Qi et al. 1999).

We found that between 14 and 64% of NPNFP-ir cells in layer V of atlantogenatans were non-pyramidal, with variation among species and neocortical region. The high frequency of atypical non-pyramidal neuron morphologies observed in the xenarthrans and afrotherians is most comparable to descriptions of the monotreme echidna neocortex based on Golgi impregnation and NPNFP immunohistochemistry (Dann and Buhl 1995; Hassiotis and Ashwell 2003; Hassiotis et al. 2005). Depending on the cortical area, between 30 and 42% of spiny neurons in echidnas exhibited an unusual non-pyramidal morphology (Hassiotis and Ashwell 2003). Similarly, Tyler and colleagues (1998) found that a large proportion of Golgi-impregnated pyramidal neurons in the visual cortex of the marsupial macropodid quokka had apical dendrites that divided into two ascending trunks. Immunostaining for NPNFP in other macropodid marsupials, the tammar wallaby (Ashwell et al. 2005) and the parma wallaby, however, yielded mostly typical pyramidal neuron morphologies in layer V. While such a high incidence of unusual projection neuron morphologies appears to be fairly common in some monotremes, marsupials, xenarthrans, and afrotherians, they are not as regularly observed throughout the neocortex in boreoeutherians (although regional variation exists—see Van der Gucht et al. 2007). For example, only 7% of Golgi impregnated neurons in somatosensory or motor cortex of rats show any atypical non-pyramidal features (Hassiotis and Ashwell 2003).

Taken together, this phylogenetic distribution suggests that while the extraverted and inverted pyramidal neuronal shapes are common across the breadth of mammalian taxa, certain clades that diverged close to the root of the mammal tree display a greater diversity of projection neuron morphologies in the neocortex. Although it remains to be determined how these non-pyramidal projection cell classes might be selective in their connectivity, it is worth noting that tract-tracing studies from rats, rabbits, and cats suggest that the inverted pyramidal neurons preferentially participate in corticocortical and corticoclaustral connections (Mendizabal-Zubiaga et al. 2007).

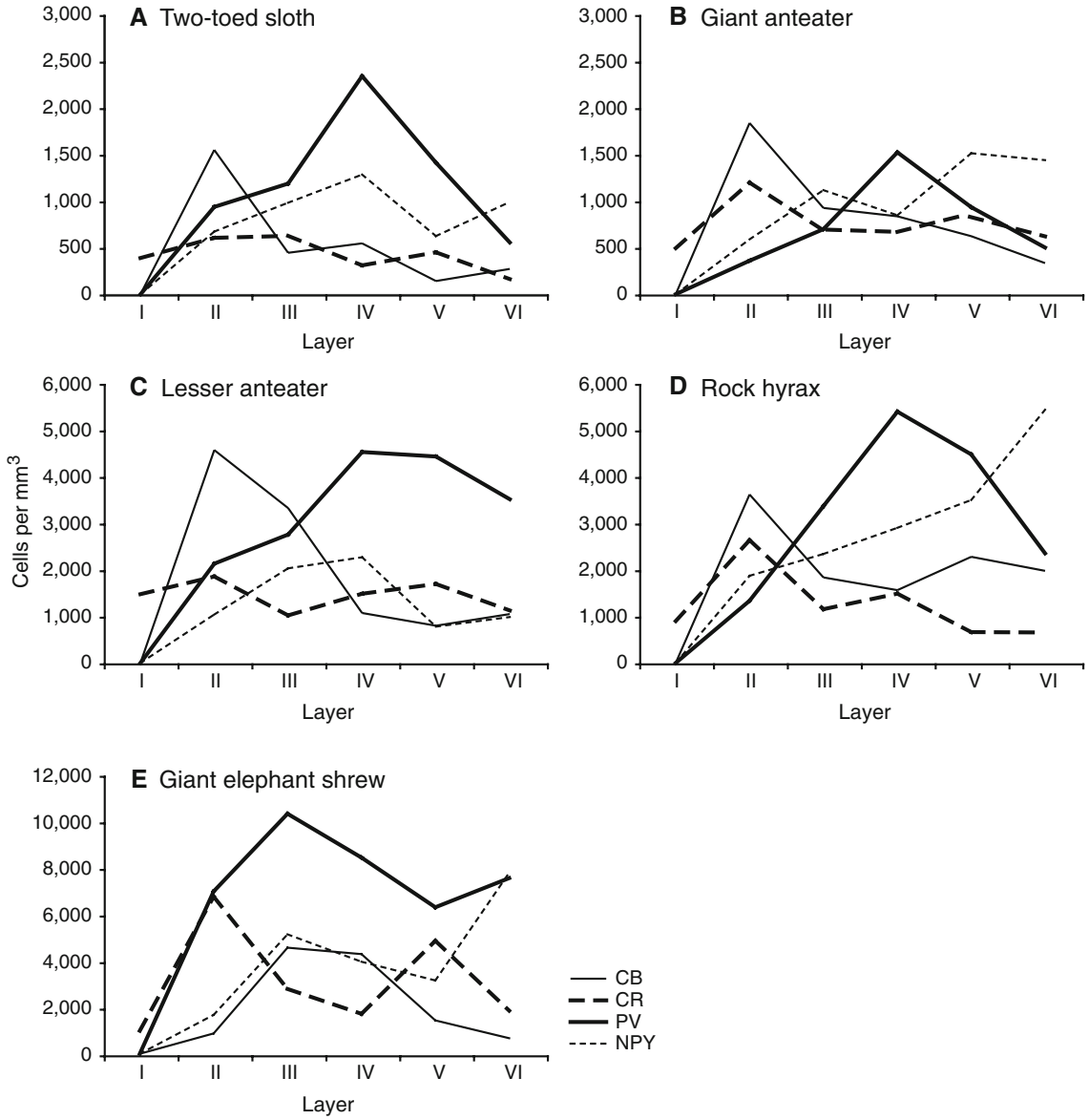


Fig. 16 Line graphs showing the relative density of interneuron subtypes in the primary somatosensory cortex. For species where more than one individual was examined, the graph represents the mean. Note that these densities are not corrected for tissue shrinkage due to fixation and histological processing; however, the densities of different subtypes within the same individual are equally affected by shrinkage

The pyramidal neurons which predominate among the NPNFP-ir cells in deep layer III and layer V in eutherians show a single long apical dendrite that reaches the upper layers, often terminating in a tuft within layer I (Boire et al. 2005; Hof et al. 1992, 1996a; Hof and Morrison 1995; Hof and Sherwood 2005; Van der Gucht et al. 2001). These apical dendrites become intertwined and form radial bundles (Feldman 1984) that are paralleled by myelinated axon fiber bundles (Peters and Sethares 1996). It is thought that this organization is an important anatomical substrate for the columnar organization of the cortex (Mountcastle 1997). However, this radial columnar architecture is not universal among mammals. Notably, the apical dendrites of the monotreme echidna appear poorly bundled in the vertical domain as revealed by NPNFP immunostaining and tangential semithin sections (Hassiotis et al. 2003, 2005). Thus, anatomical specializations for columnar modularity might have been incomplete at the origin of mammals, with further refinements added later in evolution.

In this context, it is interesting that concomitant with diversity in projection neuron morphologies, the atlantogenatans in the current study also showed clear evidence of apical dendritic bundling in NPNFP-stained sections. A Golgi-Cox staining study of the xenarthran nine-banded

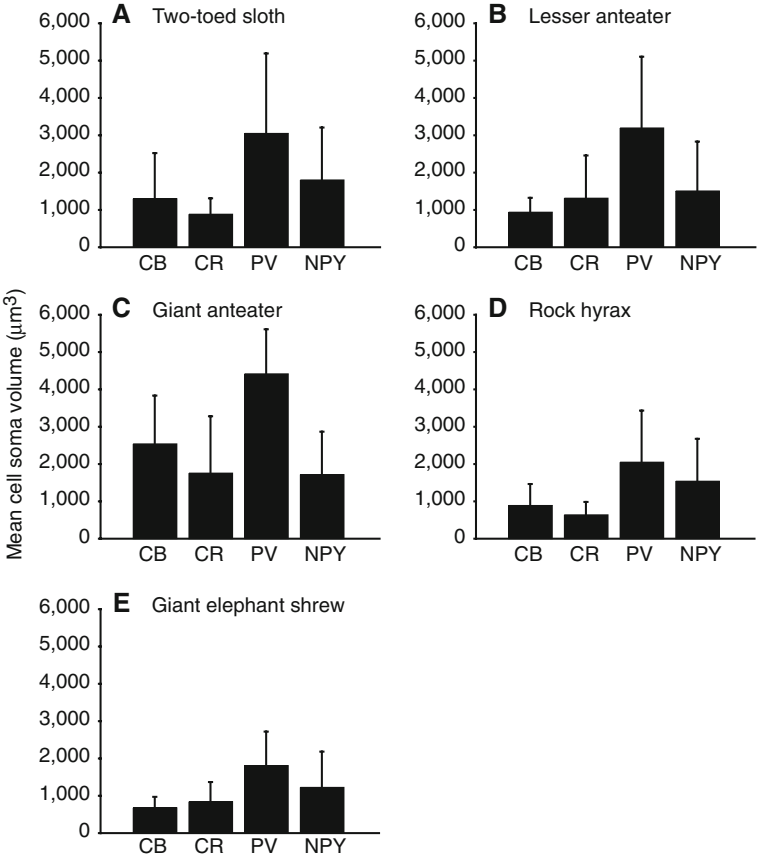


Fig. 17 *Bar graphs* showing the soma volumes of interneuron subtypes in the primary somatosensory cortex. Because initial analyses did not reveal a significant difference in cell type-specific soma volumes between individuals of the same species, data were pooled in cases where more than one individual was available for a species. Note that these volumes are not corrected for tissue shrinkage due to fixation and histological processing; however, the volumes of different subtypes within the same individual are equally affected by shrinkage. *Error bars* represent standard deviation

armadillo neocortex similarly revealed distinct apical dendrites projecting upwards in a radial orientation (Royce et al. 1975). Furthermore, Schmolke and Künzle (1997) demonstrated apical dendritic bundles arising from neurons in layers III, V, and VI in the neocortex of the afrotherian Madagascan lesser hedgehog tenrec using MAP2 immunostaining and tangential semithin sections. These data suggest that dendritic bundling might be architecturally dissociated from a regularized dominance of pyramidal-shaped projection neurons in the neocortex. We propose a scenario in which the earliest mammals had a large variety of projection neuron morphologies and comparatively little vertical cohesion to the orientation of neuronal dendrites. This stage is retained in monotremes, such as the echidna. With the origin of the first eutherian mammals, there was selection for neocortical computations involving a greater degree of interlaminar integration within minicolumn subunits, resulting in a closer spatial association among apical dendritic arbors. In these species, however, a large amount of diversity was retained in the morphology of projection neuron types. This stage is represented in Atlantogenata. The Boreoeutheria branch of mammals is more derived in displaying a higher frequency of pyramidal neuron morphologies. Comparative studies of the morphology of GFAP-ir astroglia, moreover, suggest even

Table 2 Significant Bonferroni post-hoc contrasts for the ANOVA of interneuron volumes in the primary somatosensory cortex

	CB	CR	PV	NPY
CB			**	
CR			**	**
PV	**	**		**
NPY		**	**	

** $P < 0.05$

further selection pressure for a vertical anatomical orientation to cellular elements exclusively in primates (Colombo 1996; Colombo et al. 1998, 2000, 2004).

Morphology and distribution of interneurons

The cortical GABAergic interneurons are a heterogeneous group of cells that vary in their morphology, synaptic terminations, physiology, and gene expression (for review see Ascoli et al. 2008; DeFelipe 1997; Markram et al. 2004). At the physiological level, GABA-mediated circuits participate in setting the size, structure, and response properties of receptive fields in sensory, motor, and cognitive domains (Chowdhury and Rasmusson 2002; Constantinidis et al. 2002; Fuzessery and Hall 1996; Jacobs

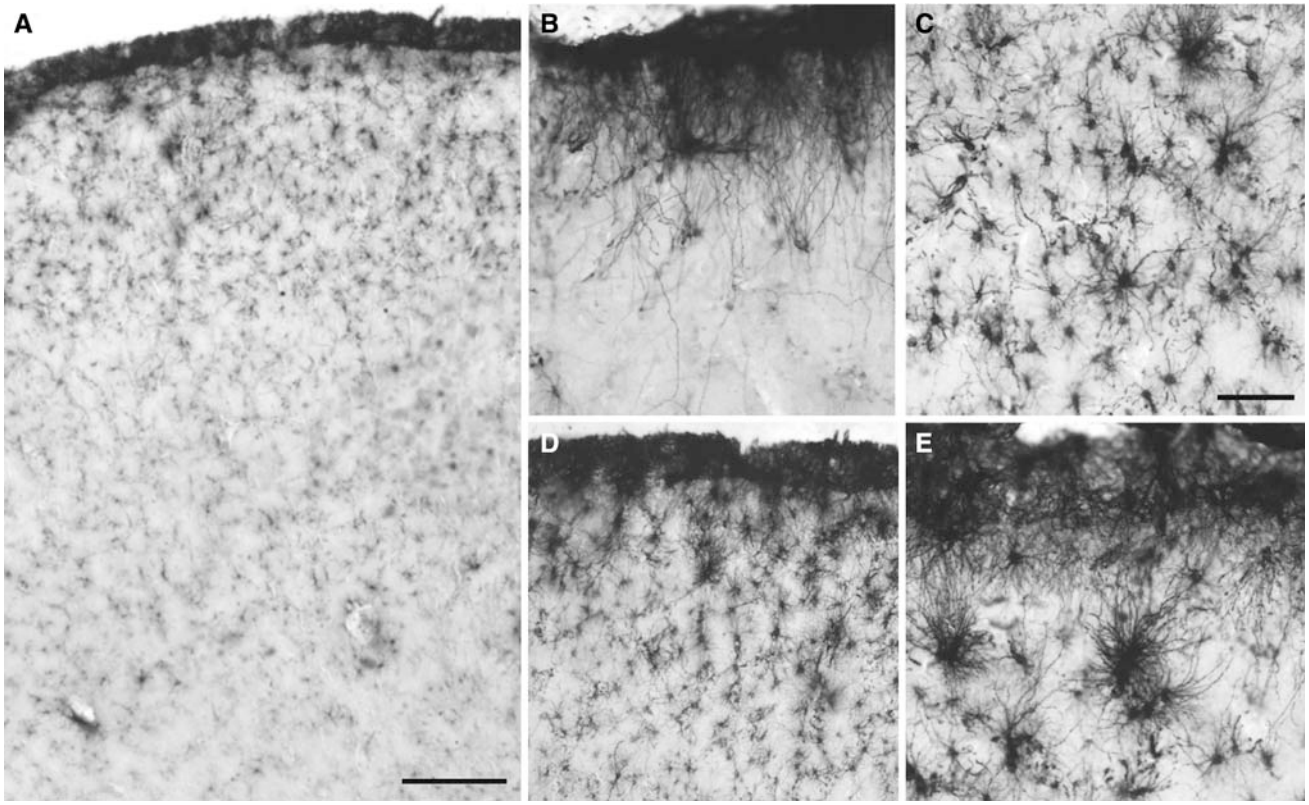


Fig. 18 Photomicrographs of GFAP immunostaining in the neocortex of various atlantogenatan species. **a** Insular cortex of a two-toed sloth. **b** Layers I–III of the dorsal frontal cortex in a giant anteater. **c** Layer III of the anterior cingulate cortex in a two-toed sloth. **d** Layers I–III of the primary auditory cortex in a rock hyrax. **e** Layers I–III of the primary somatosensory cortex in a lesser anteater. *Scale bar* in **a**, **b**, and **d** 100 μm. *Scale bar* in **c** and **e** 50 μm

Table 3 Data used in the principal components analysis

Specimen	% CB-ir interneurons	% CR-ir interneurons	% PV-ir interneurons	% NPY-ir interneurons	% NPNFP-ir neurons in layer V	% NPNFP-ir non-pyramidal neurons in layer V
Two-toed sloth 1 (TS1)	1.70	1.25	2.80	2.66	12.5	42.9
Two-toed sloth 2 (TS2)	3.10	2.72	6.50	3.32	14.1	35.9
Lesser anteater (LA)	4.33	2.99	7.19	3.12	31.4	44.9
Giant anteater (GA)	3.84	3.07	2.71	3.20	44.7	44.3
Rock hyrax 1 (RH1)	3.08	2.82	8.42	4.34	28.3	24.1
Rock hyrax 2 (RH2)	2.13	1.89	2.79	7.08	23.2	20.5
Giant elephant shrew (GES)	1.58	2.71	5.89	3.38	29.0	35.8

and Donoghue 1991; Li et al. 2002; Rao et al. 1999, 2000; Richter et al. 1999; Sillito et al. 1980; Wang et al. 2000). Because interneurons provide critical inhibitory control over the activity of projection neurons, functioning in the entrainment of ensembles of projection neurons and fine-tuning the gain of particular excitatory inputs (Markram et al. 2004), it is plausible that small modifications in this circuitry can result in profound and specific changes in cortical function (Hashimoto et al. 2003; Lewis et al. 2001).

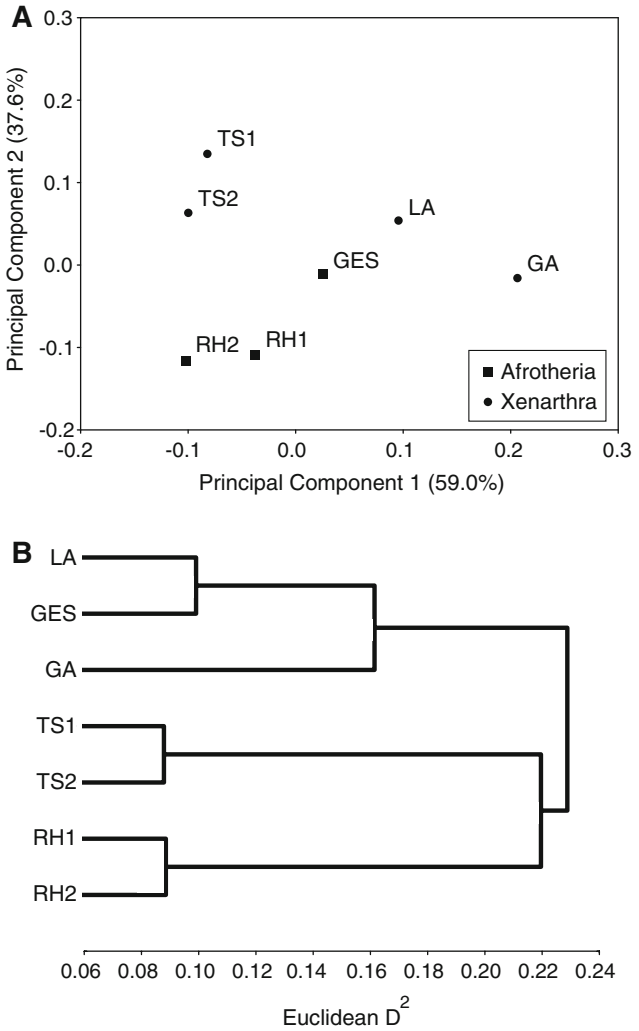


Fig. 19 Multivariate analysis of neocortical chemoarchitecture in Xenarthra and Afrotheria. **a** A plot of principal components and **b** UPGMA hierarchical clustering analysis based on the squared Euclidean distances demonstrate that phylogenetic relationships are not concordant with phenetic similarities among species. Specimen abbreviations are from Table 3

Table 4 Factor loadings from the principal components analysis

Variable	Factor 1	Factor 2
% CB-ir interneurons	0.591	−0.197
% CR-ir interneurons	0.547	−0.431
% PV-ir interneurons	−0.072	−0.134
% NPY-ir interneurons	−0.394	−0.777
% NPNFP-ir neurons in layer V	0.908	−0.419
% NPNFP-ir non-pyramidal neurons in layer V	0.589	0.808
Percent of variance	59.0%	37.6%

In a number of respects, the complement of GABAergic interneuron types appears to be common to all mammals. Many of the same morphological classes of aspiny and sparsely-spiny neocortical interneurons have been described for various monotremes, marsupials, and eutherians (Fairén et al. 1984; Hassiotis and Ashwell 2003; Tyler et al. 1998), indicating that these interneuron subtypes are universal elements of the cortical circuitry. It has been hypothesized that this diversity of inhibitory interneuron classes has arisen as an economical wiring solution to organize global synchrony and oscillations within neocortical circuits (Buzsaki et al. 2004).

The characteristics of interneuron circuitry have been studied by examining subsets of GABAergic neurons that express various calcium-binding proteins and neuropeptides (DeFelipe 1993; Markram et al. 2004). In particular, it has been established that 90–95% of GABAergic interneurons in the mammalian neocortex colocalize one of the calcium-binding proteins, CB, CR, or PV, in largely non-overlapping populations (Blümcke et al. 1990; Celio 1990; DeFelipe 1993, 1997; DeFelipe and Jones 1985; del Rio and DeFelipe 1996; Glezer et al. 1998; Hendry and Jones 1991; Hendry et al. 1989; Kosaka et al. 1987).

The morphological classes of interneurons defined by their expression of calcium-binding proteins and NPY in our study of atlantogenatans closely resembled the major types that have been described extensively in other eutherian mammals (Alcantara and Ferrer 1994, 1995; Clemo et al. 2003; Desgent et al. 2005; Gao et al. 2000; Glezer et al. 1992, 1993, 1998; Gonchar and Burkhalter 1997; Hof et al. 1996a, 1999; Hogan and Berman 1994; Ichida et al. 2000; Lund and Lewis 1993; Nimchinsky et al. 1997; Park et al. 2000; Van Brederode et al. 1990, 1991), as well as marsupials (Hof et al., 1999), and the monotreme echidna (Hassiotis et al. 2005). This suggests that the common ancestor of all mammals possessed many of the same interneuron types expressing similar neurochemical profiles and that these cell classes were retained in all descendants.

The axonal terminations of certain interneuron types also displayed similarities between the atlantogenatans and other mammals. In primates, rodents, and carnivores, subsets of large PV-ir interneurons have been identified as comprising chandelier cells and basket cells. Chandelier neurons establish terminations on the axon initial segment of postsynaptic neurons, whereas basket neurons form synapses on the soma and proximal dendrites (Kawaguchi and Kubota 1997; Somogyi et al. 1998). In the atlantogenatan neocortex, we observed short vertical PV-ir cartridges of boutons throughout layers II–IV. Similar PV-ir cartridges of synapses on the axon initial segment have been demonstrated in primates, rodents, and carnivores in superficial cortical layers (DeFelipe and Gonzalez-Albo 1998; DeFelipe et al. 1989b; del Rio and DeFelipe 1997; Erickson and Lewis 2002; Hardwick et al. 2005; Inda et al. 2008; Lewis and Lund 1990; Somogyi et al. 1985; Williams et al. 1992; Woo et al. 1998). Because a single

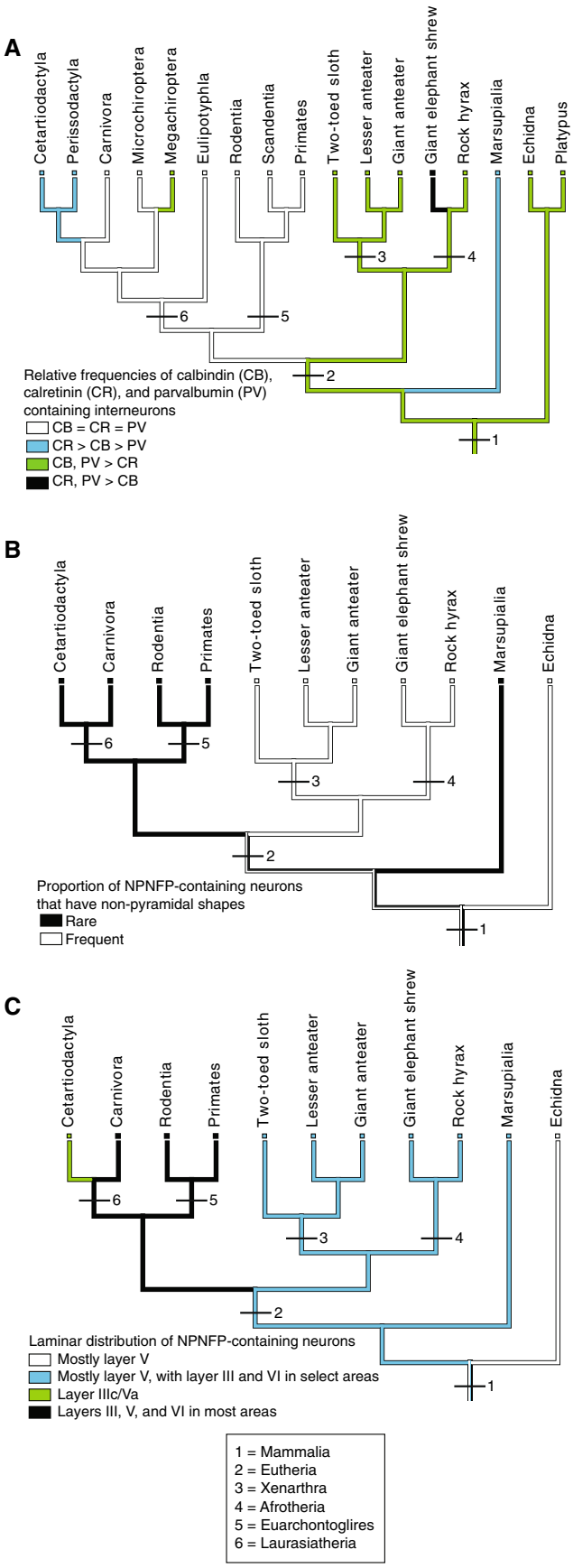


Fig. 20 The evolution of neocortical chemoarchitecture in mammals. ▶ Parsimony reconstructions of internal nodes and character evolution are shown for different traits (**a**–**c**)

chandelier cell may synapse on hundreds of local neurons, the PV-ir cartridges are thought to contribute to the synchronization of large groups of projection neurons (Somogyi et al. 1985). Interestingly, PV-ir cartridges were not observed in a study of echidna neocortex (Hassiotis et al. 2005), suggesting that certain neurochemical specializations of chandelier terminals might have arisen more recently in the eutherian branch of mammals.

We did not observe CB-ir double bouquet axon bundles in the atlantogenatans. Our findings are congruent with Ballesteros-Yañez and colleagues (2005), who found that CB-ir double bouquet axon bundles only occur in the neocortex of primates and carnivores, but not rodents, lagomorphs or artiodactyls. These long vertical axon bundles form synapses onto the dendritic spines and shafts of pyramidal neurons within a narrow column (DeFelipe et al. 1989a, 1990; Peters and Sethares 1997; Somogyi and Cowey 1981) and it has been suggested that this architecture represents a specialization of the microcolumnar organization (Ballesteros-Yañez et al. 2005).

In general among the atlantogenatans, CB- and PV-ir interneurons were more common than CR-ir interneurons throughout the neocortex. The giant elephant shrew differed, however, in displaying a relatively higher density of CR-ir interneurons. Because the monotremes at the root of the mammalian radiation are also distinguished by relatively few neocortical CR-ir interneurons, our phylogenetic analyses found that this character state in atlantogenatans is likely a conservative retention from the stem mammal, whereas the interneuron system shows more derived characteristics among marsupials and other eutherian branches of the tree. The proportional representation of calcium-binding protein-containing interneuron classes in the neocortex also revealed a range of phylogenetic variation among crown taxa within each major lineage. This suggests that there is considerable evolutionary and developmental flexibility in the determination of the phenotype of the neocortical interneuron system. Our exploratory multivariate analysis of quantitative variation in neuronal type proportions in primary somatosensory cortex of atlantogenatans, furthermore, did not group taxonomic allies together on PC1, although separation among xenarthrans and afrotherians was evident on PC2. Thus, while broad phylogenetic clades can be differentiated according to some cyto- and chemoarchitectural features of the neocortex (Hof and Sherwood 2005), there is a considerable amount of individual- and species-specific variation in the quantitative distribution of neuron types.

Table 5 Reconstructed ancestral character states

	Relative frequency of calcium-binding protein-containing interneurons		Proportion of non-pyramidal NPNFP-ir neuron morphologies		Laminar distribution of NPNFP-ir neurons	
	Parsimony	P.L.	Parsimony	P.L.	Parsimony	P.L.
Eutheria	CB, PV > CR	0.722	Frequent[a]	0.713[b]	V, with some in III and VI	0.355[b]
Xenarthra	CB, PV > CR	0.991	Frequent	0.998	V, with some in III and VI	0.997
Afrotheria	CB, PV > CR	0.843	Frequent	0.999	V, with some in III and VI	0.999
Euarchontoglires	CB = CR = PV	0.982	Rare	0.978	III, V, VI	0.969
Laurasiatheria	CB = CR = PV	0.802	Rare	0.978	III, V, VI	0.631

P.L. proportional likelihood

[a] Equivocal

[b] Not significant

These results challenge the proposition that there is a canonical inhibitory interneuron network that is relatively invariant across species and cortical areas, as has been previously suggested (Clemo et al. 2003; Douglas and Martin 2004; Hendry et al. 1987; Nelson et al. 2006; Silberberg et al. 2002). In fact, many authors have commented on phylogenetic diversity in the distribution of neocortical inhibitory interneurons among mammalian species (Ballesteros-Yañez et al. 2005; DeFelipe et al. 2002; Glezer et al. 1993; Hendry and Carder 1993; Hof et al. 1999; Hof and Sherwood 2005; Preuss and Coleman 2002; Sherwood et al. 2004, 2007). Specifically, comparative quantitative studies indicate that the overall percentage of GABAergic interneurons within the neocortex is greater in cetaceans than in rodents and primates (Glezer et al. 1993; Hof et al. 2000), and even greater still in primates compared to rodents (DeFelipe et al. 2002; del Rio and DeFelipe 1996; Gabbott and Bacon 1996; Gabbott et al. 1997; Gonchar and Burkhalter 1997). Species also vary in the proportions of different calcium-binding protein-containing interneuron subtypes within homologous neocortical areas (Gabbott and Bacon 1996; Gabbott et al. 1997; Hof et al. 1999; Sherwood et al. 2004, 2007) and the extent of their colocalization within neurons (Condé et al. 1994; Kawaguchi and Kubota 1997; Kubota et al. 1994). Finally, species also differ in the electrophysiological properties of particular interneuron classes, such as neurogliaform and basket neurons (Povysheva et al. 2007, 2008).

This range of phenotypic divergence of the interneuron system may be due to a number of factors, including Darwinian selection on genes important to the transcriptional regulation of interneuron specification and differentiation (Anderson et al. 1999; Wonders and Anderson 2006). Furthermore, it is possible that evolutionary changes in the morphology and activity of projection neurons could have an impact on the development of microcircuitry (Alpar et al. 2004).

Conclusions

Neocortical evolution has been alternately linked to the behavioral adaptations of the earliest mammalian forerunners for more precise processing of incoming sensory information (Allman 1990) or to providing a highly scalable wiring framework for expansion in the size of the forebrain (Striedter 2005). The vast majority of neuroscientific research on the neocortex, however, is based on a very small number of carnivore, rodent, and primate species. Generalizations regarding the fundamental architectural and functional features associated with the origin of mammalian neocortex require a broad comparative perspective. Our findings highlight the mosaic nature of neocortical evolution, demonstrating that particular anatomical and biochemical characteristics have accrued in different lineages of mammals.

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